Difference of Lunar Regolith Thermal Conductivity K_d at the Apollo 15 and 17 Heat-Flow Boreholes

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June 15, 2026

Abstract

The Apollo 15 and 17 landing sites are the only locations on the Moon where subsurface regolith temperatures have been measured in situ. The restored 1971–1977 Heat-Flow Experiment (HFE) record provides the sole direct constraint on the meter-scale regolith thermal conductivity, K_d, yet current thermal models apply a single Moon-wide value, for example in the calibration of orbital Diviner brightness-temperature retrievals (Hayne et al., 2017; Martínez and Siegler, 2021). We retrieve K_d separately at each borehole from the restored record (Nagihara et al., 2018) by sweeping K_d against the meter-scale-sensor RMSE (80–139 cm at Apollo 15; 130–243 cm at Apollo 17) under the Hayne et al. (2017) form, using automatically selected per-sensor stability windows and excluding borestem-contaminated shallow sensors *a priori*. Conditional on the published basal heat fluxes (21 and 15 mW m^{-2}; Langseth et al., 1976), the retrieval yields $K^*_{d,\text{A15}} = 4.58^{+1.33}_{-0.28}$ and $K^*_{d,\text{A17}} = 8.12^{+0.49}_{-0.61}$ mW m^{-1} K^{-1} (1σ statistical, non-parametric bootstrap; systematic terms in Table 4), reducing the meter-scale-sensor RMSE from 1.03 and 0.69 K under the global value $K_d = 3.4$ to 0.95 and 0.34 K. The inter-site contrast, $\Delta K^*_d = 3.3^{+0.7}_{-1.3}$ mW m^{-1} K^{-1} (95% CI [0.4, 4.6]), excludes zero at the published basal heat fluxes ($p \approx 0.01$), although propagating the basal-flux reanalysis envelope dilutes the ordering probability to ∼83%. Both retrieved values lie within porosity, composition, and orbital benchmarks. The per-site K^*_d values supply the meter-scale temperature-profile boundary condition required by future subsurface radiative-transfer retrievals, including the TSUKIMI terahertz mission, and an in-situ reference for future global conductivity models.

Plain Language Summary. *The Moon's interior is insulated from the daily surface heating by the compacted regolith (broken rock and dust) in roughly the top meter. The numbers describing that insulation come from formulas fit to satellite measurements of the lunar surface temperature, not to any direct sub-surface data. Sub-surface temperatures were measured at only two locations, by thermometers emplaced up to 1.4 m (Apollo 15) and 2.4 m (Apollo 17) deep by the Apollo 15 and 17 astronauts in 1971–1972, and the complete records were recently recovered from the original mission tapes. We use those two records to determine, separately at each borehole, the regolith conductivity in the meter-scale subsurface (80–139 cm for Apollo 15 and 130–243 cm for Apollo 17), where the daily surface temperature swing has died away and the conductivity has settled to a steady value. The Hayne (2017) global value misses the meter-scale equilibrium temperatures by about 1.0 K at Apollo 15 and 0.7 K at Apollo 17; site-by-site refitting reduces those errors, halving the Apollo 17 mismatch, and the two sites disagree on the meter-scale conductivity by roughly a factor of 1.8. The site-fit values provide the meter-scale temperature input that upcoming lunar sub-surface mapping missions will need to interpret what their instruments observe through the regolith.*

Key Points

- Meter-scale regolith conductivity K_d is retrieved per site from the restored Apollo 15 and 17 heat-flow records (80–234 cm).
- Apollo 15 gives $K_d = 4.58^{+1.33}_{-0.28}$ and Apollo 17 gives $8.12^{+0.49}_{-0.61}$ mW m^{-1} K^{-1} (1σ bootstrap).
- At published basal heat fluxes a single Moon-wide K_d is rejected ($p \approx 0.01$); meter-scale models need site-specific values.

1 Introduction

Lunar sub-surface thermal modelling presently relies on globally averaged conductivity parameterizations. We use the term *meter-scale regolith* throughout to mean the subsurface layer constrained by the Apollo HFE temperature sensors used in this retrieval—specifically, 80–139 cm for Apollo 15 and 130–243 cm for Apollo 17. At these depths the daily and annual thermal skins ($\lesssim 10$ cm and ~ 50 cm respectively) have damped to negligible amplitude, and the conductivity in both standard parameterizations has converged on its depth-independent asymptote K_d; under the Hayne (2017) form, the same K_d value extrapolates to all greater depths within the regolith column. The meter-scale regolith conductivity is supplied by formulations calibrated against orbital Diviner brightness temperatures: the smooth-exponential $K(T, z)$ of Hayne et al. (2017), in which a single meter-scale asymptote K_d is fit to a global average and applied moon-wide, and the temperature- and density-dependent $K(T, \rho)$ of Martínez and Siegler (2021), which extends the same functional family with additional physical content but is itself calibrated globally and applied uniformly. Both parameterizations probe only the diurnal skin. The extent to which the resulting global K_d represents any individual landing site, and the magnitude of any site-to-site departure, cannot be determined from orbital data alone.

Direct in-situ data exist only from the Apollo Heat-Flow Experiment (HFE). At Apollo 15 and 17, fibreglass borestems emplaced by the astronauts carried thermometers to depths exceeding two meters (Langseth et al., 1972, 1976); these two boreholes remain the only lunar locations at which a sub-surface temperature profile has been measured (Fig. 1). The restored 1971–1977 HFE record (Nagihara et al., 2018) provides the data used here. Each borehole also carries a published basal heat flux (Langseth et al., 1976; Nagihara et al., 2018) which, together with the local K_d, sets the meter-scale temperature gradient. The site-specific inputs (latitudes, albedos, emissivities, basal fluxes) are collected with their original sources in Table 1.

We retrieve K_d separately at each site by holding the Hayne et al. (2017) functional form fixed and sweeping K_d alone against the meter-scale-sensor RMSE. The retrieval is placed in context by two additional forward evaluations against the same meter-scale-sensor set: the Hayne et al. (2017) form at its published global K_d, and the Martínez and Siegler (2021) $K(T, \rho)$ model at its published coefficients (Table 2). Neither published model is refit here, so the Martínez and Siegler (2021) comparison quantifies the performance of the published global $K(T, \rho)$ at each Apollo site without per-site freedom. A full per-site retrieval under the Martínez and Siegler (2021) form requires a different choice of free parameter from K_d and is deferred to a follow-on study (§4.3).

The per-site K_d^* values matter for any sub-surface radiative-transfer (RTM) retrieval that observes the Moon at wavelengths penetrating below the diurnal skin: future microwaveand terahertz sub-surface sounders all need a meter-scale temperature column $T(z)$ as a forward-model boundary condition that orbital instruments cannot measure directly. One concrete example is the upcoming TSUKIMI mission (Lunar Terahertz Surveyor for Kilometer-scale Mapping, Kasai et al., 2022), whose forward model requires the sub-diurnal-skin temperature column that only the in-situ K_d can anchor.

Below the diurnal skin the modelled column is controlled by K_d through both the rectified annual mean and the steady gradient Q_b/K_d, so the $T(z)$ ingested by any such retrieval inherits its K_d dependence; a factor-of-two error in K_d propagates through the meter-scale temperature profile and into any retrieval that uses it. The values reported here, evaluated in the Hayne et al. (2017) thermal column, supply that boundary condition from the only in-situ data available.

The retrieval is driven by an idealized synodic insolation cycle evaluated at each site's latitude (§2); because only stability-window mean temperatures enter the fit, sub-cycle timing plays no role, and the neglected ephemeris terms (solar declination, orbital eccentricity) perturb the cycle-mean insolation at or below the percent level (§2). The remaining sections present the data and methods (§2), the per-site retrieval with the headline RMSE table (§3), the interpretation with its uncertainty limits (§4), and the conclusions.

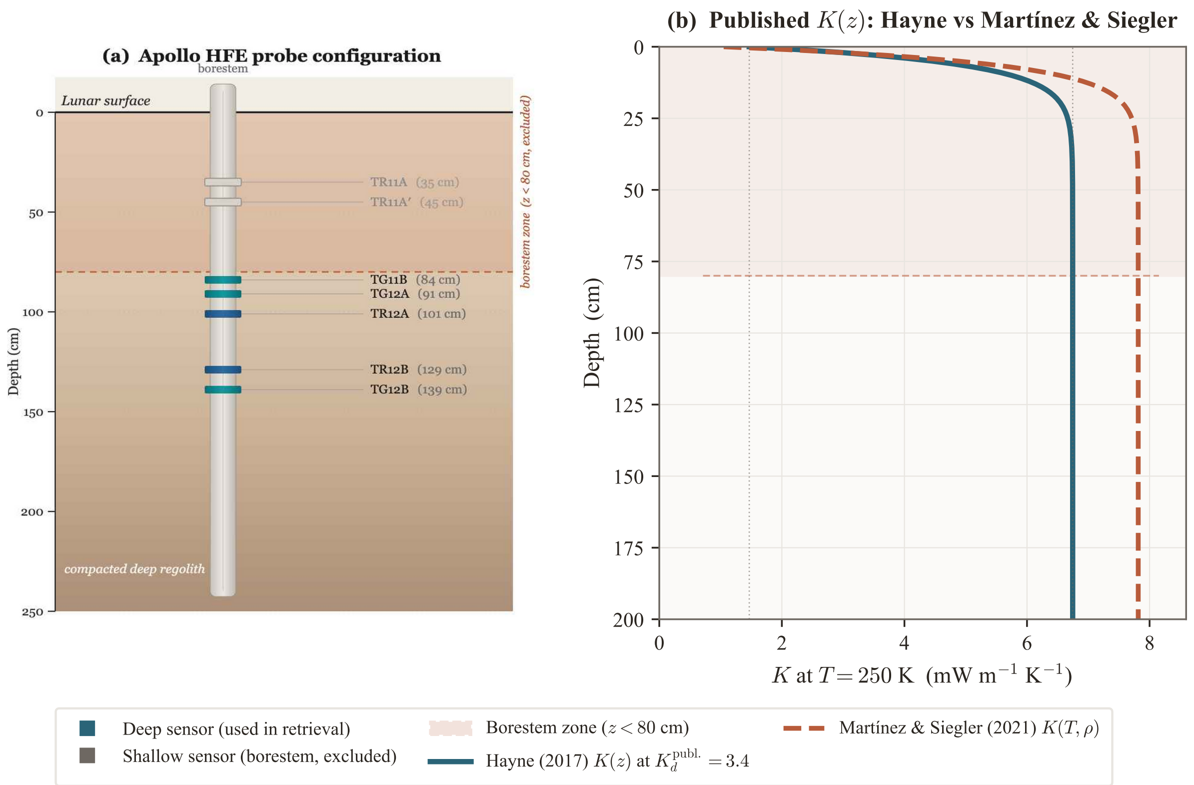

Figure 1. Apollo HFE measurement geometry and the two published $K(z)$ forms tested. **(a)** Probe schematic showing the fibreglass borestem and thermometer placements (Apollo 15 Probe-1). Coral band: borestem-excluded zone ($z < 80$ cm); teal markers: meter-scale sensors used in the retrieval. **(b)** Thermal conductivity profiles $K(z)$ at $T = 250$ K: Hayne et al. (2017) smooth-exponential form at $K_d^{\text{publ.}} = 3.4$ mW m^{-1} K^{-1} (solid teal); Martínez and Siegler (2021) $K(T, \rho)$ at published coefficients (dashed red). Both use the same Hayne-form $\rho(z)$.

2 Data and Methods

2.1 Apollo HFE meter-scale-sensor temperatures

The retrieval requires one equilibrium temperature T_{eq} per sensor. The restored 1971–1977 record (Nagihara et al., 2018) provides this quantity once the early portion of each record, which contains residual transients from drilling and emplacement, documented mission-operations disturbances (Nagihara et al., 2018), and slow instrumental drift, is excluded. For each sensor we extract automatically the longest trailing segment that is statistically indistinguishable from a flat steady state and average within it.

For each sensor, candidate start points covering 55–80% of the record (13 equally-spaced candidates; those retaining less than 20% of the record are excluded by a minimum-tail guard, capping the effective range at 80%) are evaluated by ordinary-least-squares linear fit to all temperatures between the candidate point and the end of the record. The accepted *stability window* is the earliest candidate for which the linear-fit slope satisfies $|d\langle T\rangle/dt| < 0.08\,\text{K/yr}$ ($\approx 2.2 \times 10^{-4}\,\text{K/d}$): the longest tail consistent with thermal equilibrium at this stringency. The adopted threshold is approximately 4.6× stricter than the $10^{-3}\,\text{K/d}$ value sometimes used in HFE analyses, and was chosen to suppress bias from residual post-disturbance drift. If no candidate satisfies the criterion, the last 25% of the record is used as a fallback. We report the window-mean equilibrium temperature $T_{\text{eq}} = \langle T(z)\rangle_{\Delta t}$ together with its within-window standard deviation, which serves as the per-sensor measurement uncertainty in downstream statistics. Sensors with central depth $z < 80\,\text{cm}$ (the borestem-contaminated zone; §2.4) are plotted but excluded from the RMSE. Figure 3 shows the full per-probe temperature record at each site, with the documented disturbance intervals shaded (Nagihara et al., 2018), the selected stability windows marked, and the fitted long-term drift $d\langle T\rangle/dt$ reported per sensor.

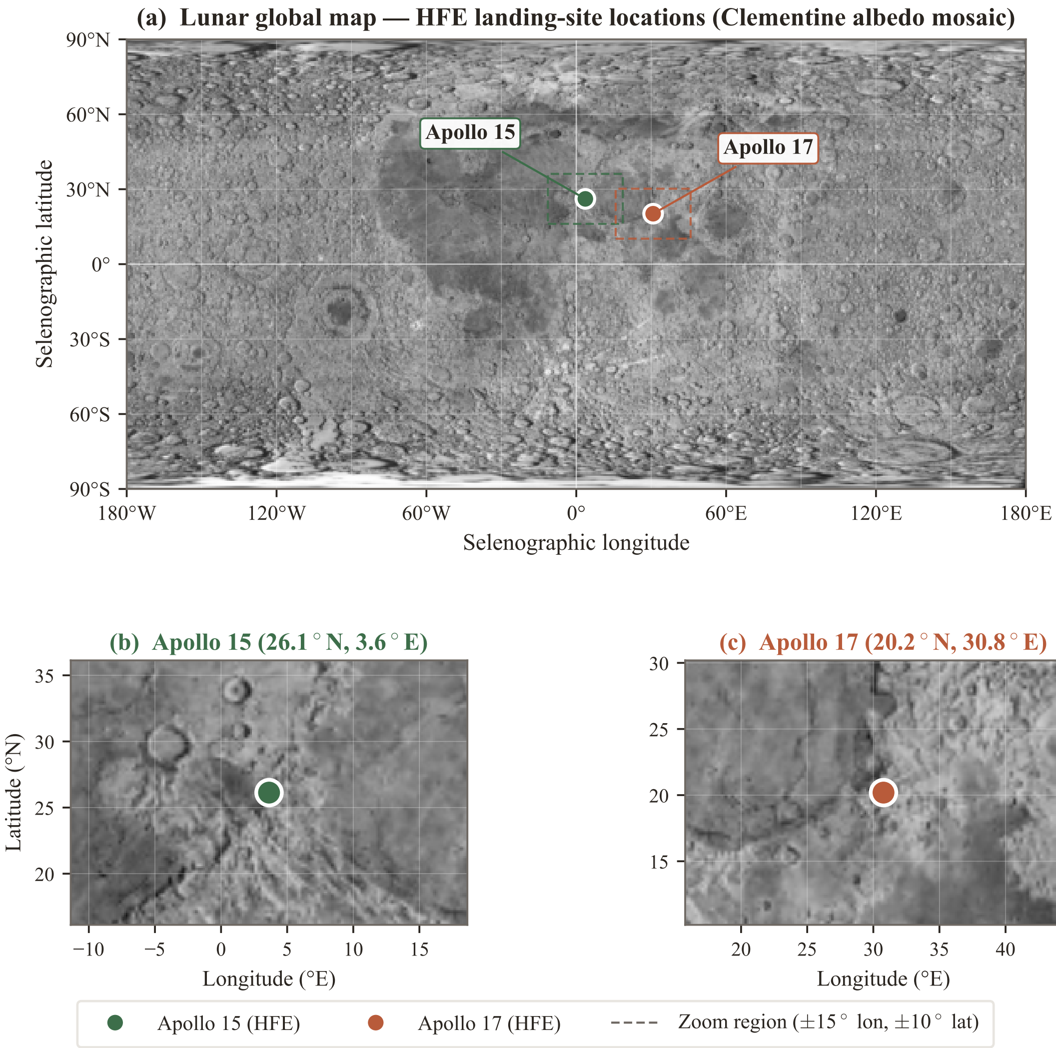

Figure 2. Apollo Heat-Flow Experiment landing-site context on the Clementine global albedo mosaic (USGS/NASA, public domain). **(a)** Global equirectangular map showing the two HFE boreholes; dashed coloured rectangles mark the zoom regions of panels (b) and (c). **(b)** Apollo 15 regional zoom (±15° longitude, ±10° latitude), Imbrium–Apennine boundary, mare/highland margin. **(c)** Apollo 17 regional zoom, basaltic embayment of Mare Serenitatis at the Taurus–Littrow highlands edge (Korotev, 2005; Wieczorek et al., 2006; Crawford, 2015). Borehole coordinates from the ALSEP gazetteer.

The retrieval is insensitive to the precise threshold. Sweeping the threshold over 0.04–0.16 K/yr (a factor of four around the adopted value) shifts K_d^* by at most 0.02 mW/(m K) at A15 (0.4% of nominal) and 0.05 mW/(m K) at A17 (0.6%), in both cases an order of magnitude below the bootstrap 1σ uncertainty (§3.2). The adopted threshold therefore lies within the flat band of the threshold response.

2.2 Surface solar geometry and incident insolation

The solver is forced with an idealized synodic insolation cycle: the daytime solar zenith angle at each site is approximated by $\cos\theta_\odot(t) = \cos\phi\,\cos(2\pi t/P)$, with ϕ the site latitude and $P = 29.531$ d the mean synodic period. Because the retrieval compares stability-window *mean* temperatures (§2) rather than time-resolved phases, ephemeris-level timing corrections do not enter the fit. The neglected solar declination (±1.54°) and orbital eccentricity (±3.3% peak-to-peak in instantaneous flux, < 0.1% in the cycle mean) perturb the mean insolation at or below the percent level, comparable to and subsumed by the Bond-albedo sensitivity of Table 4. The Moon has no atmosphere, so the solar irradiance incident on the local horizontal surface is

$$F_\odot(t) = \max\big(0,\ S_0\cos\theta_\odot(t)\big), \tag{1}$$

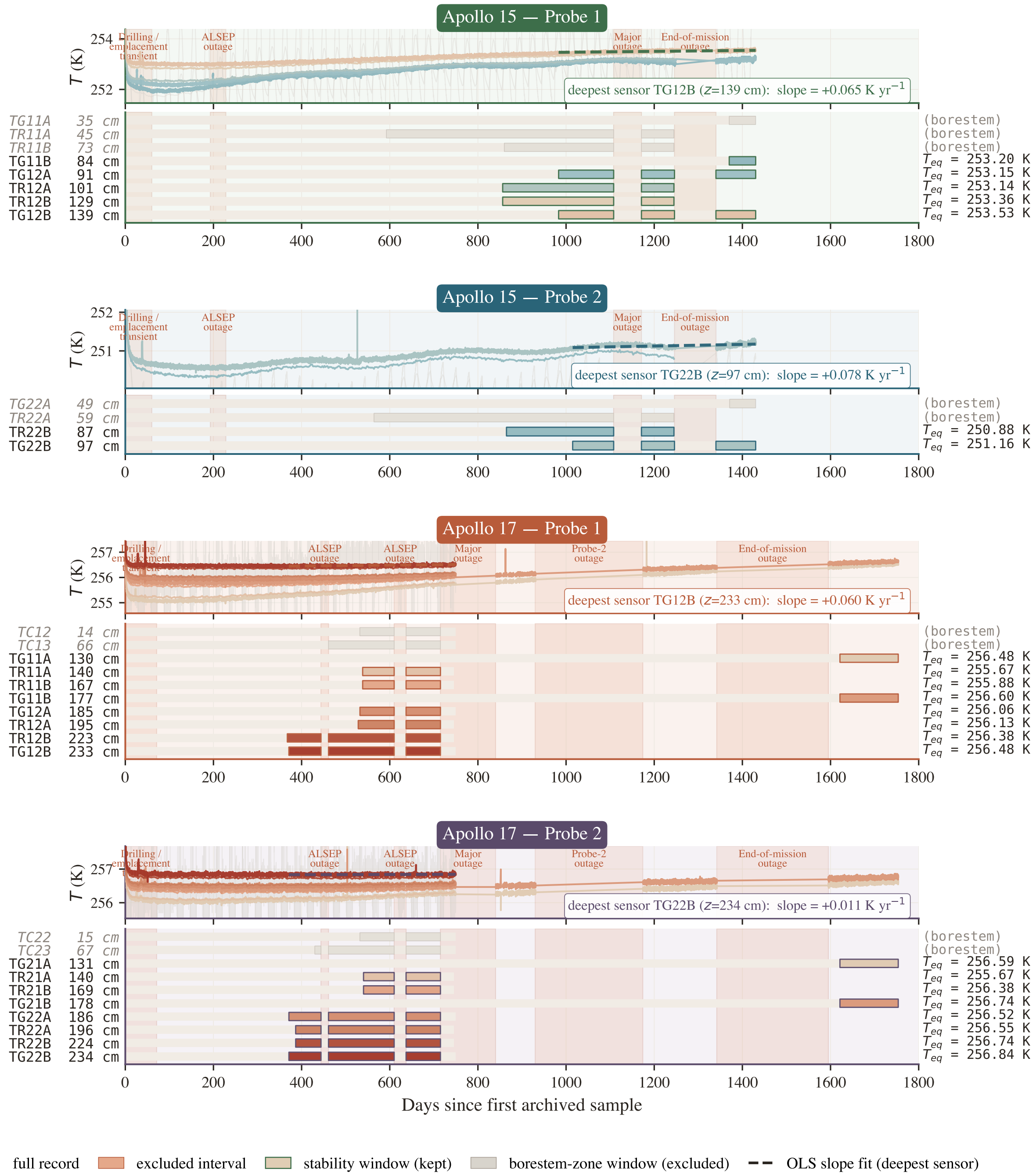

Figure 3. Per-probe stability-window timeline at Apollo 15 and 17 (4 probes). Each probe panel shows, above, the raw HFE traces with coral bands marking documented data-quality events and the selected stability windows outlined; and, below, a depth-coded per-sensor Gantt strip broken at outages. Selection criteria and the role of the window-mean T_{eq} are described in §2.

with $S_0 = 1361\,\mathrm{W/m^2}$ (Kopp and Lean, 2011).

2.3 1-D solver and Hayne et al. (2017) thermophysics

The depth coordinate z is defined positive downward, with the regolith surface at $z = 0$ and the lower boundary of the modelled column at $z = z_{\max} = 5$ m. Conservation of energy in one dimension gives the heat equation

$$\rho(z)\, c_p(T)\, \frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\Big[K(T, z)\, \frac{\partial T}{\partial z} \Big]. \tag{2}$$

At the upper boundary the skin temperature T_s satisfies radiative balance between absorbed insolation, emitted thermal radiation, and the conductive flux into the column,

$$(1-A)\,F_{\odot}(t) \;-\; \varepsilon\sigma T_s^4 \;-\; \left(-K\,\partial_z T\right)_{z=0} \;=\; 0, \tag{3}$$

with $\sigma = 5.6704\times10^{-8}$ W m^{-2} K^{-4} the Stefan–Boltzmann constant; equation (3) is solved iteratively for T_s at each time step. At the lower boundary a constant geothermal flux is imposed,

$$-K\,\partial_z T\big|_{z_{\max}} = Q_b. \tag{4}$$

The PDE is discretised on a geometric grid ($\Delta z_0 = 2$ mm, growth ratio 1.08) and integrated with a Crank–Nicolson scheme at $\Delta t = 1$ h. The periodic steady state is computed by a flux-anchored outer iteration rather than a fixed-length spin-up: short multi-lunation integrations converge the diurnal skin, after which the sub-skin mean profile is reconstructed from the steady-state mean-flux closure $d\langle T\rangle/dz = [Q_b - u_{\rm rect}(z)]/K(\langle T\rangle, z)$, where the rectified (eddy-correlation) flux $u_{\rm rect} = \langle K\,\partial_z T\rangle - K(\langle T\rangle)\,\partial_z\langle T\rangle$ is measured from the resolved cycle; the two steps are alternated until the anchor temperature at $z = 0.55$ m is stable to 5×10^{-3} K. This construction is required because a flux-bottom-boundary column relaxes on the diffusive timescale of the full 5-m domain ($\sim10^3$ lunations), so no fixed-length spin-up of practical cost is free of its initial condition. Convergence is certified rather than assumed: initial temperature guesses of 240 and 260 K yield mean profiles agreeing to ≤ 0.03 K everywhere in the column, and a 120-lunation verification integration started from the converged profile drifts by < 0.09 K at the sensor depths, bounding the solver systematic on K_d^* at $\lesssim 0.15$ mW m^{-1} K^{-1} — an order of magnitude below the bootstrap uncertainty (§3.2). The Hayne et al. (2017) thermophysical inputs are adopted as given in Appendix A of the original paper:

$$K(T,z) = \Big\{K_s + (K_d - K_s)\big[1 - e^{-z/H}\big]\Big\}\cdot\Big[1 + \chi\,(T/T_{\rm ref})^3\Big], \tag{5}$$

$$\rho(z) = \rho_d - (\rho_d - \rho_s)\,e^{-z/H}, \tag{6}$$

$$c_p(T) = c_0 + c_1T + c_2T^2 + c_3T^3 + c_4T^4. \tag{7}$$

Numerical values and sources for every coefficient, together with the site-specific measured quantities (latitude, albedo, emissivity, basal heat flux), are listed in Table 1. The basal heat flux reanalyses of Saito et al. (2007); Nagihara et al. (2018) are folded into the sensitivity sweep of §3.2.

As an independent comparison we also evaluate the Martínez and Siegler (2021) regolith conductivity model. Where Hayne et al. (2017) prescribes a depth-dependent envelope through $K(T,z) = [K_s + (K_d - K_s)(1 - e^{-z/H})](1 + \chi(T/T_{\rm ref})^3)$, Martínez and Siegler (2021) replaces that envelope with an explicitly density-dependent two-mechanism decomposition,

$$K(T,\rho) = \underbrace{(A_1\rho + A_2)\,k_{\rm am}(T)}_{\text{contact (phonon)}} \;+\; \underbrace{(B_1\rho + B_2)\,T^3}_{\text{radiative}}. \tag{8}$$

The first term is the contact (phonon) conductivity transmitted through the network of grain-to-grain contacts; it carries the temperature dependence of an amorphous solid through the low-temperature polynomial $k_{\rm am}(T)$ of Martínez and Siegler (2021) and scales linearly with the local bulk density, reflecting the increased number of grain-to-grain contacts per unit volume at higher packing. The second term is the radiative contribution from infrared photon exchange across pore voids; it scales as T^3 (the derivative of the Stefan–Boltzmann emission with respect to temperature, characteristic of photon-mediated transport) and is density-modulated with the opposite physical content, since denser packing reduces the void fraction available for radiative exchange. The four coefficients (A_1, A_2, B_1, B_2) are calibrated globally against Diviner brightness temperatures and are listed in Table 1. The density profile $\rho(z)$ used in this comparison is the same Hayne-form exponential used in the Hayne column, so the two models differ in their K formulation alone.

The Martínez and Siegler (2021) formulation has no single K_d analogous to the parameter retrieved under the Hayne form: both terms in equation (8) vary continuously with depth through

Table 1. Fixed inputs to the 1-D heat solver. Three blocks: (A) the Hayne et al. (2017) conductivity-form coefficients (eq. 5), with K_d the single parameter retrieved in §2.5; (B) the four published coefficients of the Martínez and Siegler (2021) $K(T, \rho)$ comparison model (eq. 8, forward only in the headline table; see also §4.3); (C) site-specific measured quantities shared by both conductivity models. The shared $\rho(z)$ uses the Hayne-form profile (eq. 6); the specific heat is the Hemingway et al. (1973) polynomial $c_p(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4$ with $(c_0, \ldots, c_4) = (-3.61,\ 2.74,\ 2.36 \times 10^{-3},\ -1.23 \times 10^{-5},\ 8.91 \times 10^{-9})$.

Parameter	Symbol	Value	Unit	Source
(A) Hayne et al. (2017) smooth-exponential conductivity, all values from Hayne et al. (2017) App. A unless noted				
Surface conductivity	K_s	7.4×10^{-4}	$\mathrm{W\,m^{-1}\,K^{-1}}$	
Deep conductivity (global)	K_d (publ.)	3.4×10^{-3}	$\mathrm{W\,m^{-1}\,K^{-1}}$	
Deep conductivity (per-site)	K_d^*	§3.2	$\mathrm{W\,m^{-1}\,K^{-1}}$	**this work**
$K(z)$ transition scale	H	0.06	m	
Radiative coefficient	χ	2.7	—	
Radiative normalisation	T_{ref}	350	K	
(B) Martínez & Siegler (2021) $K(T, \rho)$ (forward only)*, coefficients from Martínez and Siegler (2021)*				
Contact, ρ-slope	A_1	5.082×10^{-6}	$\mathrm{W\,m^{-1}\,K^{-1}\,(kg\,m^{-3})^{-1}}$	
Contact, intercept	A_2	-5.10×10^{-3}	$\mathrm{W\,m^{-1}\,K^{-1}}$	
Radiative, ρ-slope	B_1	2.002×10^{-13}	$\mathrm{W\,m^{-1}\,K^{-4}\,(kg\,m^{-3})^{-1}}$	
Radiative, intercept	B_2	-1.953×10^{-10}	$\mathrm{W\,m^{-1}\,K^{-4}}$	
Shared by both conductivity models				
Surface bulk density	ρ_s	1100	$\mathrm{kg\,m^{-3}}$	Hayne et al. (2017)
Deep bulk density	ρ_d	1800	$\mathrm{kg\,m^{-3}}$	Hayne et al. (2017)
Specific heat	$c_p(T)$	polyn. (caption)	$\mathrm{J\,kg^{-1}\,K^{-1}}$	Hemingway et al. (1973)
(C) Site-specific measured inputs (A15 / A17)				
Latitude	ϕ	26.1° / 20.2° N	—	ALSEP gazetteer
Bond albedo	A	0.131 / 0.137	—	Vasavada et al. (2012)
Broadband emissivity	ε	0.95	—	Bandfield et al. (2015)
Basal heat flux	Q_b	21 / 15	$\mathrm{mW\,m^{-2}}$	Langseth et al. (1976); Nagihara et al. (2018)
Solar constant	S_0	1361	$\mathrm{W\,m^{-2}}$	Kopp and Lean (2011)

$\rho(z)$ and with temperature through $T(z)$. The published model contains no free parameter, so its evaluation at each site yields a single forward prediction (Table 2). To diagnose, in the discussion (§4.3), whether the published Martínez and Siegler (2021) form could fit both sites if it were granted one per-site lever analogous to the Hayne K_d, we additionally perform an alternative retrieval in which the asymptote of the Hayne-form density profile entering equation (8) is rescaled by a single per-site scalar α,

$$\rho_d^{\mathrm{site}} \;=\; \alpha \cdot 1800\ \mathrm{kg\,m^{-3}}, \tag{9}$$

with $\rho_s = 1100$ kg m^{-3}, the e-folding depth H, and the published coefficients (A_1, A_2, B_1, B_2) and $k_{\mathrm{am}}(T)$ held fixed; $\alpha = 1$ recovers the published forward prediction. The scalar α is our diagnostic construction (the Martínez and Siegler (2021) model itself is not refit, only re-evaluated at a per-site density), and its physical interpretation is the per-site asymptotic bulk density that the published $K(T, \rho)$ would require to reproduce the in-situ data. The α retrieval is not part of the headline K_d result; it is interpreted in §4.3.

2.4 Borestem diagnostic and the meter-scale-sensor cut

The meter-scale ($z > 80$ cm) sensor set used in the K_d retrieval is defined by a physics-based amplitude diagnostic, not by the residuals it would produce. The amplitude of the diurnal wave serves as a frequency-domain check independent of the annual-mean comparison. The per-sensor diurnal amplitude is estimated as $\sqrt{2}\,\sigma_{\Delta t}$, where $\sigma_{\Delta t}$ is the within-stability-window standard deviation; the proxy equals the half-range exactly for a sinusoidal signal, is diurnal-dominated for shallow sensors, and saturates at the digitisation noise floor below the skin depth — precisely the contrast the diagnostic requires (Fig. 4). Two features are diagnostic. Above $z = 80$ cm, observed amplitudes exceed the modelled regolith-attenuation curve by an order of magnitude or more, the signature of heat conducted along the fibreglass borestem from the lunar surface, which thermally short-circuits the upper $\sim 80\,\mathrm{cm}$ of regolith (Nagihara et al., 2018). Below 80 cm, both published $K(z)$ forms predict diurnal amplitudes $\lesssim 10^{-7}$ K ($\sim$18 e-foldings of the diurnal skin depth $\delta \sim 3$–5 cm), two to three orders of magnitude below the 0.05–0.2 K digitisation noise floor of the restored record. Meter-scale-sensor amplitudes therefore do not constrain the thermal diffusivity $\kappa = K/(\rho c_p)$ at the in-situ scale; the Hayne and Martínez and Siegler (2021) attenuation curves plotted in Figure 4 agree to within $\sim$10% in skin depth and overlap visually below the noise floor, both indicating "deep = unresolved". The shallow-sensor amplitude excess, evaluated against a prediction that depends only on $K(T, z)$, $\rho(z)$, and $c_p(T)$, defines the meter-scale ($z > 80\,\mathrm{cm}$) validation set used in the retrieval.

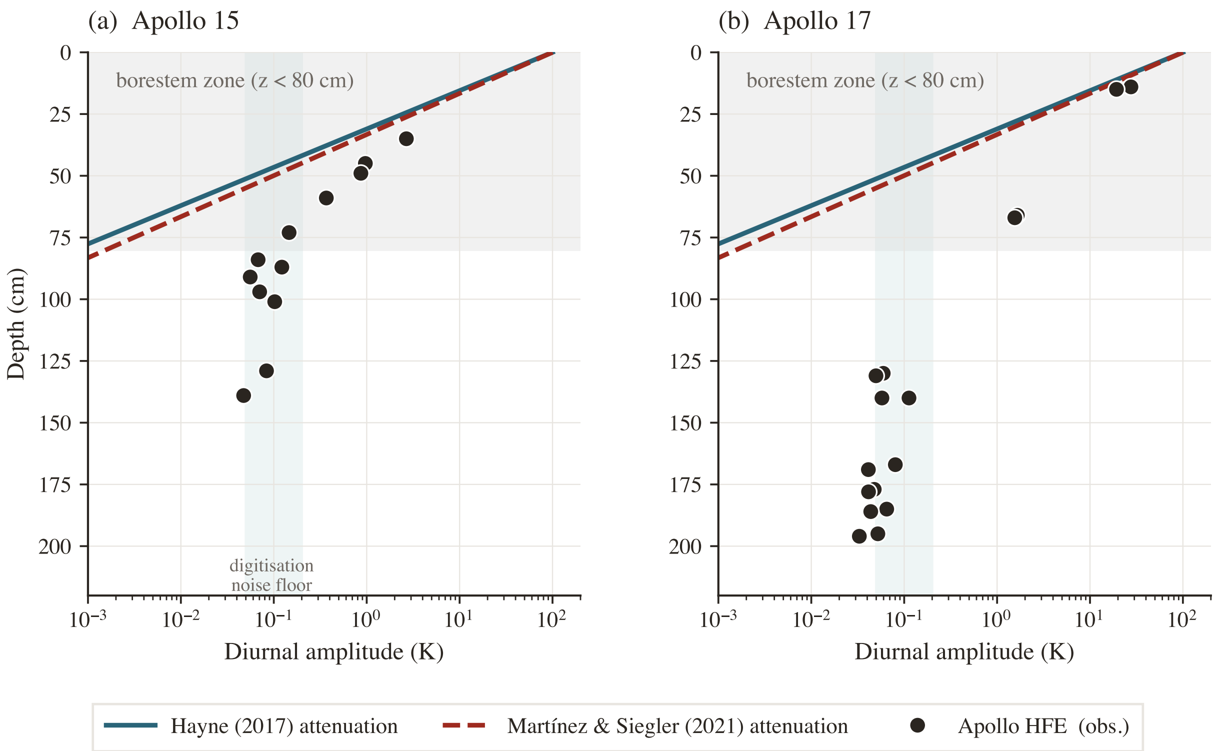

Figure 4. Diurnal amplitude (estimated as $\sqrt{2}\times$ the within-stability-window standard deviation, the half-range of a sinusoidal signal) versus sensor depth at both Apollo sites. Black markers: observed; solid teal: Hayne et al. (2017) regolith attenuation; dashed red: Martínez and Siegler (2021) attenuation, both evaluated at $T = 250$ K with the published meter-scale regolith density of 1800 kg m^{-3}and $c_p(250\,\mathrm{K}) \approx 670$ J kg^{-1} K^{-1}, the Hayne form including its radiative conductivity term. The two model curves differ by $\sim$10% in predicted skin depth and are visually indistinguishable on the log axis at all relevant depths. The shaded band is the borestem zone ($z < 80$ cm). The order-of-magnitude shallow-sensor excess is the borestem heat-short signature.

2.5 Per-site K_d retrieval (Hayne shape held fixed)

K_d is retrieved separately at each site by minimising the meter-scale-sensor RMSE while holding the remainder of the Hayne functional form (eqs. 5–7) fixed. $(K_s, H, \chi, T_{\mathrm{ref}})$ and the density and c_p functions are held at their published values, and K_d is swept across site-specific grids: 28 points spanning 1.0–15.0×10^{-3} W/(m K) at A15 and 30 points spanning 3.0–25.0×10^{-3} W/(m K) at A17, sized so that both parabolic minima and the bootstrap upper percentiles fall inside the swept range (Fig. 7). At each K_d a full Crank–Nicolson spin-up is run to convergence, and the meter-scale-sensor RMSE is evaluated as

$$\mathrm{RMSE}(K_d) = \left\langle \left[T_{\mathrm{model}}(z_i; K_d) - T_{\mathrm{eq}}^{\mathrm{HFE}}(z_i)\right]^2 \right\rangle^{1/2}_{z_i \geq 80\,\mathrm{cm}}. \tag{10}$$

The point estimate K_d^* is the parabolic minimum through the three grid values bracketing arg min RMSE. Its uncertainty is quantified by a non-parametric bootstrap with $N_{\mathrm{boot}} = 1500$ resamples of the

Table 2. Headline meter-scale-sensor RMSE table. Meter-scale-sensor (z = 80–234 cm) validation statistics for the three forward configurations confronted with the same Apollo HFE record: the Hayne et al. (2017) form at its published global K_d, the Hayne et al. (2017) form with K_d refit per site (the retrieval of §3.2), and the Martínez and Siegler (2021) $K(T, \rho)$ model evaluated forward at each site. K_d in mW m^{-1} K^{-1} (95% bootstrap CI in brackets for the Hayne site-fit row). Bias = mean(model − obs). A diagnostic per-site retrieval under the Martínez and Siegler (2021) form is presented separately in §4.3.

Site	Model	Fitted knob	N	RMSE (K)	Bias (K)
Apollo 15	Hayne (2017), global K_d	—	7	**1.03**	+0.49
Apollo 15	Hayne, K_d^* fit	$K_d^* = 4.57\,[4.12, 7.45]$	7	**0.95**	+0.32
Apollo 15	Martínez and Siegler (2021) forward	—	7	**0.99**	+0.41
Apollo 17	Hayne (2017), global K_d	—	16	**0.69**	+0.16
Apollo 17	Hayne, K_d^* fit	$K_d^* = 8.12\,[6.85, 9.04]$	16	**0.34**	+0.04
Apollo 17	Martínez and Siegler (2021) forward	—	16	**0.64**	−0.27

meter-scale-sensor set drawn with replacement, with the ±2.5 cm sensor-placement uncertainty (Nagihara et al., 2018) propagated as a per-resample Gaussian depth jitter, following standard small-sample regression practice (Press et al., 2007). The bootstrap distribution defines 95% confidence intervals on K_d^* and on the inter-site contrast ΔK_d^*, and provides the null-hypothesis test for inter-site equality (Fig. 7). Percentile-method intervals are reported throughout; bias-corrected accelerated (BCa) intervals (Efron and Tibshirani, 1993) agree with the percentile endpoints to within 1.3 mW/(m K) at both sites and on the contrast, and do not alter any qualitative conclusion. The retrieval has one free parameter.

Models with different free-parameter counts on the same meter-scale-sensor set are compared using the small-sample-corrected Akaike information criterion (Burnham and Anderson, 2002), $\mathrm{AICc} = N \ln(\mathrm{RSS}/N) + 2k + 2k(k+1)/(N-k-1)$, where RSS is the meter-scale-sensor residual sum of squares and k is the number of fitted parameters ($k = 0$ for fixed-K_d rows, $k = 1$ for each retrieval). Differences ΔAICc are reported relative to the best model at each site; following Burnham and Anderson (2002), $\Delta\mathrm{AICc} \lesssim 2$ indicates statistically indistinguishable models and $\Delta\mathrm{AICc} \gtrsim 10$ a decisive preference. A reduced χ^2 is not reported: the within-stability-window scatter ($\sigma \sim 0.03$–0.05 K) characterises short-term thermometer precision rather than the model-adequacy error budget, yields $\chi^2_\nu \gg 1$ for every model, and is therefore uninformative for model selection in this context.

3 Results

3.1 Annual-mean subsurface profile and the global-model gap

Both published global models reproduce the annual-mean borehole profiles to within ∼1 K, but systematic per-site residuals remain. Figure 5 compares the two modelled time-averaged $\langle T(z)\rangle$ profiles, Hayne et al. (2017) at the global $K_d = 3.4$ mW/(m K) and Martínez and Siegler (2021) $K(T, \rho)$ evaluated forward, against the stability-window means. The published Hayne value runs warm relative to the HFE record by a mean of +0.49 K at A15 and +0.16 K at A17, with RMSE 1.03 and 0.69 K — the residual structure, not the mean offset, carries the per-site information; the Martínez and Siegler (2021) prediction is comparable (RMSE 0.99 and 0.64 K) (Table 2). The shaded band ($z < 80$ cm) marks the borestem-contaminated zone, excluded *a priori* from the quantitative comparison (§2.4). The residual structure that remains — in particular the under-predicted temperature gradient between the 80-cm and deepest sensors at A17 — motivates the per-site K_d retrieval of §3.2.

3.2 Per-site K_d retrieval

The meter-scale-sensor RMSE exhibits a well-resolved minimum at each site:

$$K^*_{d,\mathrm{A15}} = 4.58^{+1.33}_{-0.28}\,[4.12,\ 7.45]\,\mathrm{mW/(m\,K)}, \qquad K^*_{d,\mathrm{A17}} = 8.12^{+0.49}_{-0.61}\,[6.85,\ 9.04]\,\mathrm{mW/(m\,K)},$$

with asymmetric 1σ bootstrap uncertainties and 95% confidence intervals in brackets (Fig. 7). The meter-scale-sensor RMSE decreases from 1.03 K and 0.69 K under the published global K_d to 0.95

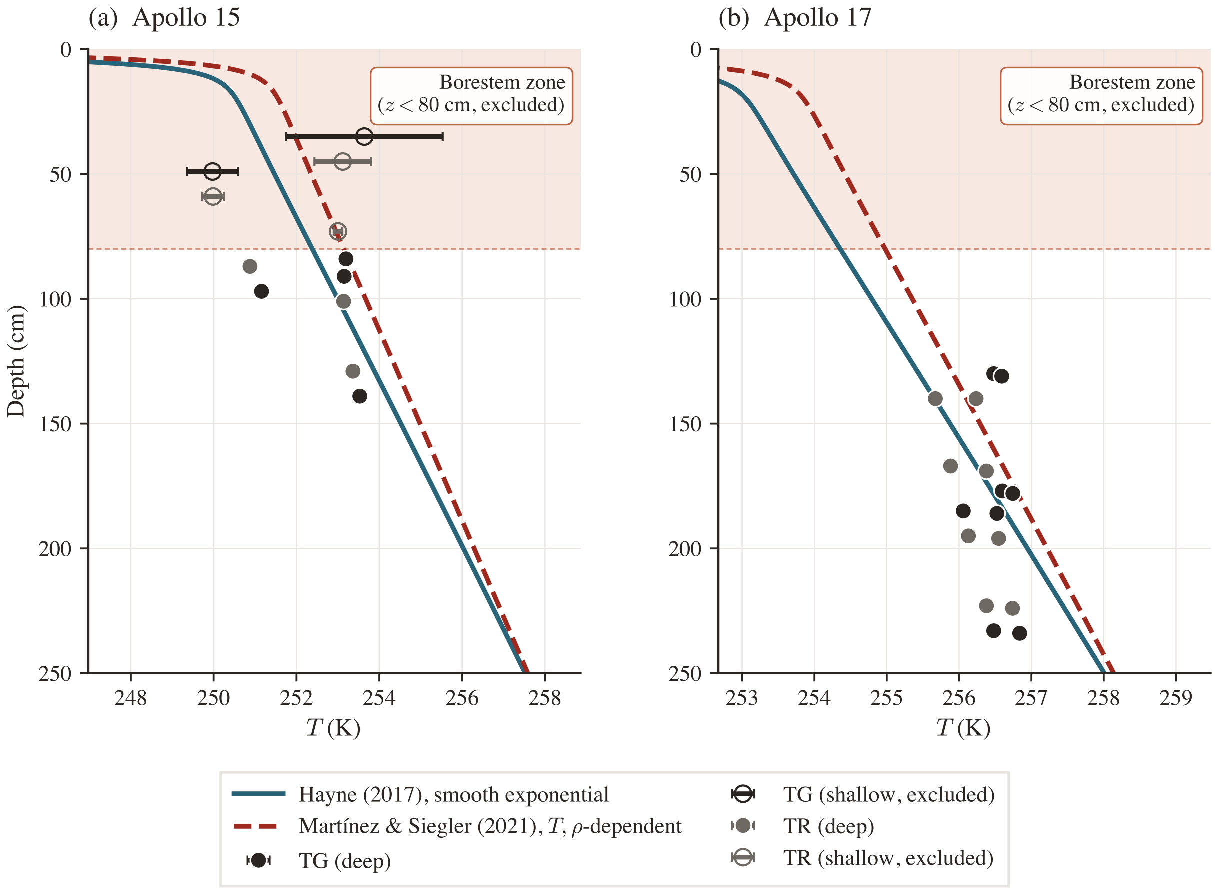

Figure 5. Annual-mean subsurface temperature $\langle T(z) \rangle$ for the two published conductivity models at their nominal (un-retrieved) parameters, Hayne et al. (2017) smooth exponential at the global $K_d = 3.4\,\mathrm{mW/(m\,K)}$ (solid), and the Martínez and Siegler (2021) temperature- and density-dependent $K(T, \rho)$ forward (dashed), against the Apollo HFE stability-window means. Markers are coded by sensor type (charcoal = TG, grey = TR); horizontal bars are the within-window standard deviation. The shaded band is the borestem-contaminated zone ($z < 80\,\mathrm{cm}$), excluded from the RMSE.

K and 0.34 K under the per-site retrieval ($N = 7$ at A15, $N = 16$ at A17; Table 2); at A17 the refit halves the RMSE and removes the gradient misfit (bias +0.04 K), while at A15 the improvement is modest and the global value is not excluded by the RMSE criterion alone — it is excluded by the bootstrap CI, whose lower bound (4.12) lies above 3.4. K_d is therefore the dominant identifiable lever between the Hayne form and the in-situ HFE record once the borestem zone is excluded, with the caveat that the per-site evidence is decisive at A17 (ΔAICc ≈ 19 against the global value) but not at A15 alone (ΔAICc ≈ 3 in favour of the *global* value once the parameter penalty is paid; the inter-site contrast, not the A15 refit, carries the *statistical* result.

The minimum is not an artefact of the sweep grid. Extending the sweep beyond the fitting envelope reveals that each RMSE curve rises monotonically on the high-K_d side (Fig. 6), confirming bracketed minima rather than descending shoulders. Nor is it an artefact of the thermal-state initialisation: the equilibrium construction of §2.3 is certified independent of its initial guess to ≤ 0.03 K, which bounds the solver contribution to K_d^* at $\lesssim 0.15\,\mathrm{mW/(m\,K)}$, an order of magnitude below the bootstrap uncertainty quoted above.

The Martínez and Siegler (2021) model, evaluated forward at each site with its published global coefficients, serves as the global-baseline comparison: meter-scale-sensor RMSE of 0.99 K at A15 and 0.64 K at A17, with mean biases of +0.41 K and −0.27 K (Table 2). It reproduces both sites to within ~1 K without any per-site freedom — comparable to the Hayne site-fit at A15, though still a factor of ~1.9 above it at A17. A diagnostic per-site retrieval under this form is presented in §4.3, where the implications for the $K(T, \rho)$ parameterization are discussed.

Sensitivity to the borestem exclusion depth. The retrieval is repeated with the borestem exclusion depth z_b swept over 60–90 cm. The retrieved values remain stable across all admissible cuts: $K^*_{d,\mathrm{A15}} = 4.6 - -4.9$ and $K^*_{d,\mathrm{A17}} = 8.1\,\mathrm{mW/(m\,K)}$ for $z_b = 70$, 80, and 90 cm, yielding a contrast of 3.2–3.6 mW/(m K), unchanged within rounding from the nominal value. The 60 cm cut alone deviates ($K^*_{d,\mathrm{A17}} = 9.1$, contrast 4.2), because at this depth a sensor inside the borestem-affected zone re-enters the A17 set and inflates the retrieved gradient, precisely the contamination that the amplitude-based cut (§2.4) is designed to exclude. The retrieval is therefore insensitive to z_b across its physically defensible range, and the 60 cm outlier confirms rather than undermines the borestem diagnostic.

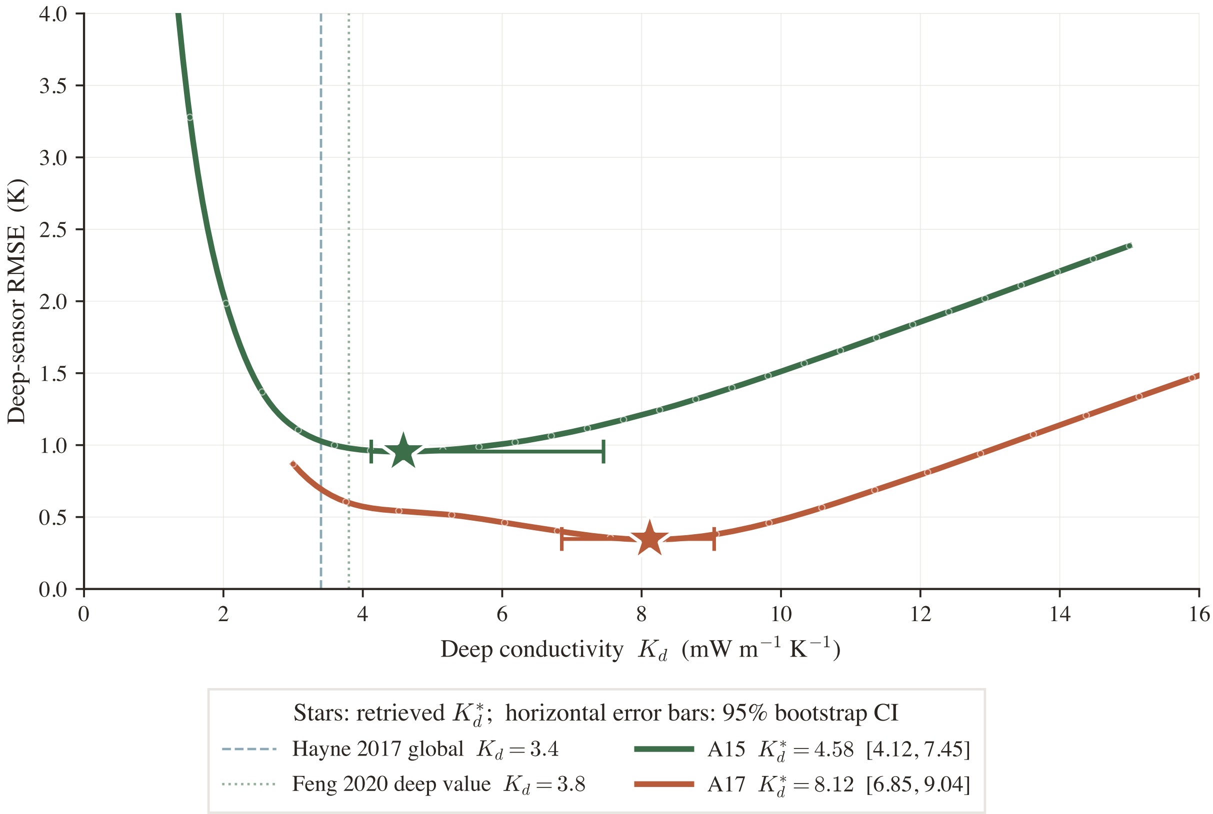

Figure 6. Per-site K_d retrieval under the Hayne et al. (2017) functional form, with all other Hayne et al. (2017) parameters held at their published values. Meter-scale-sensor RMSE(K_d) at each site, extended past the fitting grid so each curve is shown rising again on the high-K_d side: the minima are genuine bracketed bowls, not descending shoulders. Markers: forward-model evaluations; stars: retrieved K_d^*; horizontal error bars: 95% bootstrap CI. Dashed vertical: Hayne et al. (2017) global $K_d = 3.4$; dotted: the Feng et al. (2020) deep value $K_d = 3.8$.

Two summary statistics must be distinguished. The K_d^* values quoted above are parabolic point estimates of the RMSE minimum; the bootstrap medians of the same quantities are 4.63 and 8.13 mW/(m K) (Table 4). The inter-site contrast is defined as the bootstrap median of the paired within-resample difference, $\Delta K_d^* = 3.31^{+0.71}_{-1.28}\,\mathrm{mW/(m\,K)}$ (1σ, asymmetric; 95% CI [0.37, 4.58], $p \approx 0.011$ for the null of inter-site equality; Fig. 7). The paired construction preserves the resample-level correlation between sites and is therefore the statistically appropriate estimator; this is also the reason 3.31 differs slightly from the difference of the point estimates ($8.12 - 4.58 = 3.53$). The A15 upper tail reflects a weakly constrained shoulder in RMSE(K_d) at A15 for $N = 7$ meter-scale sensors, even though the central parabolic minimum is firmly at 4.58. The reported 1σ intervals propagate sensor-set composition, the ±2.5 cm depth-placement uncertainty (Nagihara et al., 2018), and the parabolic curve-fit error; component-by-component impacts of the remaining Hayne inputs are tabulated in Table 4. Both retrievals reject the global $K_d = 3.4\,\mathrm{mW/(m\,K)}$ of Hayne et al. (2017) at the 95% level(bootstrap CI lower bounds 4.12 and 6.85).

Applied at each site, the retrieved K_d^* values yield meter-scale-sensor profiles that track the HFE

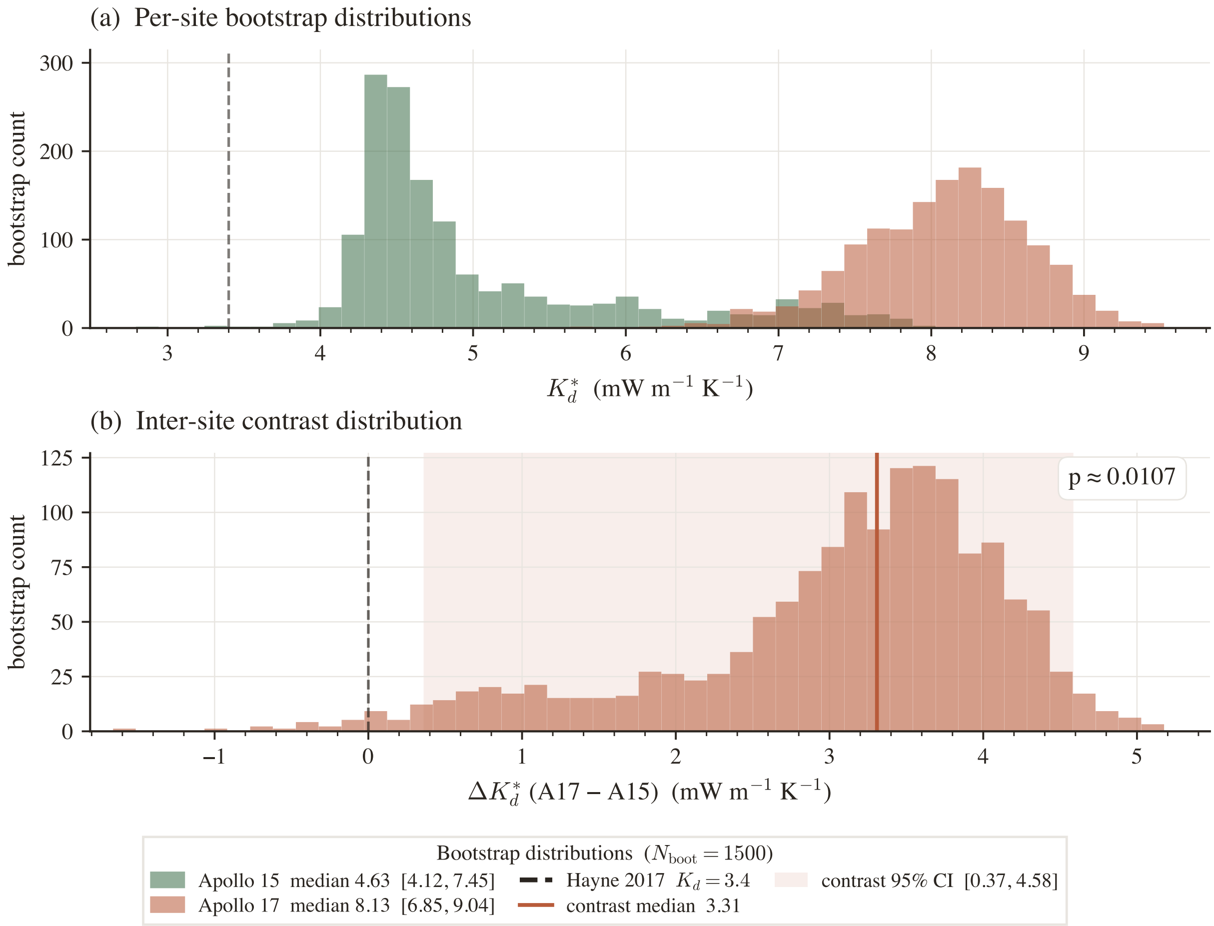

Figure 7. Bootstrap distributions ($N_{\rm boot} = 1500$, meter-scale-sensor resampling with ± 2.5 cm depth jitter). (a) Per-site K_d^* histograms with 95% CIs in the legend; dashed vertical at the Hayne et al. (2017) global value. (b) Inter-site contrast $\Delta K_d^* = K_d^{\rm A17} - K_d^{\rm A15}$; 95% CI shaded, lower bound above zero ($p \approx 0.011$).

observations within the ~ 1 K spread of the data (Fig. 8). The Martínez and Siegler (2021) forward, plotted alongside, reproduces A15 to within a factor of $\sim$1.2 of the Hayne site-fit but underpredicts A17.

A full per-component error budget for K_d^* is deferred to §4.2 (Table 4), where the basal-flux degeneracy that dominates the budget is discussed.

3.3 Consistency checks

Three diagnostics test the per-site K_d^* values. Their status as independent or internal is identified explicitly.

An independent Diviner surface-temperature closure test (Fig. 11, discussed in §4.3) shows that both conductivity models reproduce the canonical lunar diurnal shape, confirming that the diurnal forcing chain is correctly captured. Two further internal checks follow.

(i) TG vs. TR cross-prediction. Fitting K_d on the gradient-bridge sensors (TG) alone and predicting the ring-bridge (TR) sensors, and vice versa, yields held-out RMSEs within $\sim$0.26 K of the in-sample values at both sites. *(ii) Leave-one-deepest-out.* Withholding the deepest sensor ($z = 139$ cm at A15, $z = 234$ cm at A17) and refitting K_d on the remainder predicts the held-out temperature to within $|\Delta| = 0.21$ K and $|\Delta| = 0.09$ K respectively. Both internal checks confirm that the retrieval is not driven by a single sensor or sensor type; neither constitutes an independent test of the inter-site contrast.

Bayesian cross-check. The non-parametric bootstrap is reported as the primary uncertainty estimate throughout because it makes no assumption about the basal heat flux and propagates only quantities measured at the boreholes (sensor-set composition and placement depth). As an

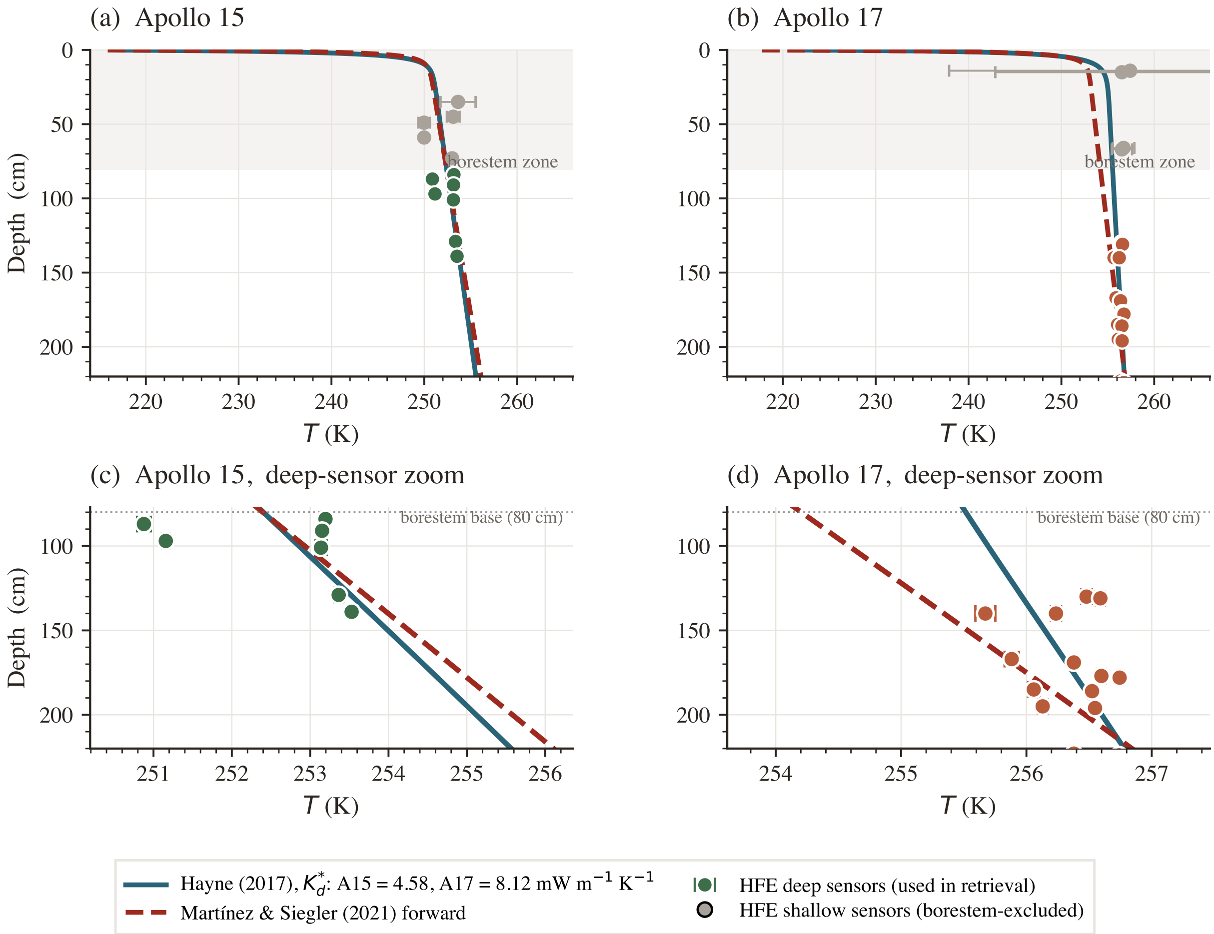

Figure 8. Annual-mean $T(z)$ at the two sites under the two conductivity models, against the Apollo HFE meter-scale sensors. Solid teal: Hayne et al. (2017) at the per-site retrieved K_d^*; dashed red: Martínez and Siegler (2021) forward (no fitted knob). Top: full profile; bottom: meter-scale-sensor zoom. Coloured markers: HFE meter-scale sensors (used in retrieval); grey markers: borestem-affected shallow sensors (excluded).

independent cross-check on the *direction* of the contrast, not as a second estimate of its magnitude, we also ran a Bayesian inversion propagating the Saito et al. (2007); Nagihara et al. (2018) prior on Q_b. Its marginal MCMC medians are $K_{d,\mathrm{A15}} = 3.6$ and $K_{d,\mathrm{A17}} = 6.1$ mW m^{-1} K^{-1} (contrast median $+2.5$). These differ from the bootstrap point estimates (4.58, 8.12; contrast $+3.3$) by construction, not by disagreement: the Q_b prior slides each marginal along the K_d/Q_b degeneracy ridge (§4.1), so the two methods are not independent measurements of the absolute K_d. The ordering survives the Q_b smearing with $P(K_d^{\mathrm{A17}} > K_d^{\mathrm{A15}}) \approx 83\%$: conditional on the published basal fluxes the contrast is significant ($p \approx 0.011$), but admitting the full reanalysis envelope on Q_b dilutes it — the honest statement is that the contrast in the meter-scale *temperature gradient* is established, while its attribution to conductivity rather than basal flux depends on the Q_b assumption (§4.2). The bootstrap interval on the contrast magnitude ($\Delta K_d^* = 3.3$, 95% CI $[0.4, 4.6]$) is carried forward as the headline measurement.

4 Discussion

4.1 The two boreholes require different meter-scale temperature gradients

The HFE record does not admit a single meter-scale conductivity that fits both sites. Under the Hayne form at the published basal heat fluxes, the value that fits the A17 meter-scale-sensor residual (8.12) degrades A15 to a +0.79 K bias (RMSE 1.22 K), and the value that fits A15 (4.58) degrades the A17 RMSE from 0.34 to 0.54 K. The published global value of Hayne et al. (2017), applied unchanged, runs warm at both sites (+0.49 K at A15, +0.16 K at A17; Table 2) and, more importantly, under-predicts the A17 meter-scale temperature gradient. The per-site retrievals are $K_d^{\mathrm{A15}} = 4.58$ and $K_d^{\mathrm{A17}} = 8.12\,\mathrm{mW/(m\,K)}$ (95% bootstrap CIs [4.12, 7.45] and [6.85, 9.04]). The lower 95% bound on the contrast distribution (Fig. 7b) lies above zero ($\Delta K_d^* = 3.31$, 95% CI [0.37, 4.58], $p \approx 0.011$): conditional on the published basal heat fluxes, both the *requirement* of different meter-scale temperature gradients and the *magnitude* of the conductivity contrast are established at the 95% level. The conditionality matters — under the K_d/Q_b degeneracy of §4.2 the attribution to conductivity rather than basal flux still rests on the Q_b assumption.

The pooled three-model comparison (Table 3) points the same way, more weakly. A per-site K_d with Q_b fixed at the published values (M1) is preferred over a shared K_d with per-site Q_b free within the published envelope (M2, $\Delta\mathrm{AICc} = 4.0$) and over the strict null of shared K_d and fixed Q_b (M3, $\Delta\mathrm{AICc} = 2.7$). Under the standard interpretation of the small-sample AIC, this constitutes only weak-to-moderate pooled evidence; the pooled statistic dilutes the A17 signal with the less-informative A15 set ($N = 7$), whereas the per-site comparison is decisive at A17 ($\Delta\mathrm{AICc} \approx 19$ for the site fit over the global value) and the paired bootstrap puts the contrast above zero at $p \approx 0.011$. The data therefore favor a per-site difference in meter-scale temperature gradient without excluding a uniform-conductivity alternative in which the two basal heat fluxes differ.

Table 3. Pooled meter-scale-sensor model comparison ($N = 23$: 7 at A15, 16 at A17). M1 retrieves a per-site K_d; M2 shares one K_d and frees the per-site Q_b within the published envelope; M3 fixes both. ΔAICc is relative to the best model. K_d in $\mathrm{mW\,m^{-1}\,K^{-1}}$, Q_b in $\mathrm{mW\,m^{-2}}$.

Model	K_d (A15 / A17)	Q_b (A15 / A17)	RMSE (K)	ΔAICc
M1 variable K_d, Q_b fixed	5.0 / 8.0	21 / 15 (fixed)	**0.60**	**0.0**
M2 uniform K_d, Q_b free	7.0 (shared)	14 / 16 (fitted)	0.61	+4.0
M3 uniform K_d, Q_b fixed	6.0 (shared)	21 / 15 (fixed)	0.67	+2.7

4.2 The retrieved absolute K_d values are degeneracy-limited

The absolute K_d^* values must be interpreted as conditional on the adopted basal heat flux, not as free-standing measurements. At steady state, the meter-scale temperature gradient $|\partial_z T| = Q_b/K_d$ is invariant under any rescaling $(K_d, Q_b) \to (\alpha K_d, \alpha Q_b)$: a single borehole cannot resolve these two products independently. The degeneracy is mapped quantitatively in Fig. 9a and is visible in the joint Bayesian posteriors. Adopting the canonical $Q_b^{\mathrm{A17}} = 15\,\mathrm{mW/m^2}$ (Langseth et al., 1976; Nagihara et al., 2018) yields $K_d = 8.12$; substituting the lower $\sim$10 mW m^{-2} value of Saito et al. (2007); Nagihara et al. (2018) rescales K_d downward to $\sim$5.4 mW m^{-1} K^{-1}, while the upper bound $\sim$18 mW m^{-2} shifts it upward to $\sim$9.7 mW m^{-1} K^{-1}. The A17 K_d^* therefore carries a $\sim^{+20}_{-33}\%$ envelope determined entirely by the basal-flux debate; A15 carries a comparable envelope. The quantity constrained tightly at each borehole is the product Q_b/K_d, equivalently the meter-scale temperature gradient itself. This represents the intrinsic limit of what two boreholes can deliver.

Reading the per-component error budget. The derivation of each row in Table 4 falls into one of three categories: σ_{stat} is the 16–84 percentile half-width of the bootstrap distribution under the $\pm$2.5 cm astronaut-emplacement envelope reported by Nagihara et al. (2018); σ_{Q_b} is obtained in closed form through the K_d/Q_b degeneracy across the published reduction range (Langseth et al., 1976 canonical; Saito et al., 2007; Nagihara et al., 2018 reanalysis); every remaining row is computed by re-running the full retrieval at the perturbed input, with the envelopes taken from the published literature (σ_A: the Vasavada et al. (2012) lat-band albedo scatter of $\pm$0.01; σ_{K_s}: the Cremers and Birkebak (1971) laboratory scatter on Apollo 11 fines; σ_ρ: the Mitchell et al. (1973); Carrier et al. (1991) Apollo-core compaction range; σ_{solver}: the certified residual of the equilibrium construction, §2.3).

How the total is combined. The **Total (quadrature)** row is the independent-errors combination $\sigma_{\mathrm{tot}} = \sqrt{\sum_i \sigma_i^2}$ over the statistical, solver, measured-input, and methodological rows. The worked arithmetic is

$$\sigma_{\mathrm{tot}}^{\mathrm{A15}} = (0.80^2 + 0.15^2 + 1.20^2 + 0.80^2 + 0.53^2 + 0.01^2 + 0.01^2 + 0.15^2)^{1/2} = 1.75,$$

Table 4. Per-component 1σ error budget for K_d^*. Values in mW m^{-1} K^{-1}; entries below 0.05 are reported as <0.05. All components are re-derived with the certified equilibrium solver. The quadrature total covers the statistical and measured-input components; χ and H parameterize the Hayne et al. (2017) functional form that the retrieval holds fixed, and are reported separately as conditionality terms (see text) rather than summed — re-retrieving at the Vasavada et al. (2012) normalisation $\chi = 1.48$ drives the RMSE minimum outside the physical sweep range at both sites.

Component	Envelope	Method	A15	A17
Statistical and solver				
$\sigma_{\rm stat}$	± 2.5 cm depth jitter	bootstrap 16–84 % half-range	0.80	0.55
$\sigma_{\rm solver}$	≤ 0.08 K residual drift	equilibrium certification bound (§2.3)	0.15	0.15
Measured inputs				
σ_{Q_b}	Q_b A15 $\in [14, 25]$, A17 $\in [10, 18]$	K_d/Q_b degeneracy mapping	1.20	2.16
σ_A (albedo)	$A \pm 0.01$	re-retrieve K_d^* at perturbed A	0.80	3.50
σ_{K_s}	$K_s \pm 30\%$	re-retrieve at perturbed K_s	0.53	1.96
σ_ρ	$\rho_d \in [1700, 2000]$ kg m^{-3}	re-retrieve at perturbed ρ_d (ρ enters ρc_p only)	<0.05	0.42
Methodological choices (this work)				
$\sigma_{\rm thr}$	0.04–0.16 K/yr (4× adopted)	threshold sweep, half-range of K_d^*	<0.05	<0.05
σ_{z_b}	$z_b \in \{70, 80, 90\}$ cm	cut sweep, half-range of K_d^*	0.15	<0.05
Total (quadrature)			**1.75**	**4.61**
Bootstrap median K_d^*			4.63	8.13
Conditionality of the held-fixed form (reported, not summed)				
χ	$\chi \in \{1.48, 2.7\}$	re-retrieve at the Vasavada et al. (2012) endpoint	minimum exits sweep (≤ 1.5 / ≥ 22)	
H	$H \in [3, 14]$ cm	per-H K_d^* half-range along the valley	0.43	5.32

$$\sigma_{\rm tot}^{\rm A17} = (0.55^2 + 0.15^2 + 2.16^2 + 3.50^2 + 1.96^2 + 0.42^2 + 0.03^2 + 0^2)^{1/2} \approx 4.61 \text{ mW m}^{-1}\,\text{K}^{-1}.$$

We flag one residual correlation: $\sigma_{\rm stat}$ and σ_{Q_b} both propagate along the K_d/Q_b degeneracy ridge of Fig. 9a, so the totals are independence-idealised rather than fully decorrelated standard errors.

The conditionality terms. The χ and H rows are deliberately excluded from the quadrature: they are not measured inputs with Gaussian envelopes but coefficients of the Hayne et al. (2017) functional form that the retrieval holds fixed. Re-retrieving at the Vasavada et al. (2012) normalisation $\chi = 1.48$ drives the RMSE minimum outside the physical sweep range at both sites — under the converged thermal physics, the rectified annual-mean temperature at sensor depths depends strongly on the radiative coefficient, so the Hayne form at the Vasavada χ cannot reproduce the HFE record at any admissible K_d. K_d^* is therefore reported conditional on the published $(\chi, H) = (2.7, 6\,\text{cm})$, in precisely the same sense as it is conditional on the published Q_b; the inter-site *contrast* is far less exposed, since both sites share the same form.

Three conclusions follow. First, the A15 budget is balanced among σ_{Q_b}, $\sigma_{\rm stat}$, and σ_A (jointly $\sim$87% of the quadrature sum), whereas the A17 budget is dominated by σ_A and σ_{K_s}: at A17 the annual-mean response to K_d is nearly flat over part of the sweep range, so even the ± 0.01 albedo envelope moves the minimum substantially — the statistical precision of the A17 retrieval (CI half-width $\sim$1.1) is real, but its accuracy is systematics-limited at the ± 4.6 level. Second, σ_ρ is structurally suppressed under the Hayne $K(T, z)$ form (eq. 5), which contains no explicit ρ dependence: $\rho(z)$ enters only the volumetric heat capacity ρc_p and perturbs the rectified annual-mean column at the $\lesssim 0.4$ mW m^{-1} K^{-1} level. Third, the methodological-choice rows $\sigma_{\rm thr}$ and σ_{z_b} sit one to two orders of magnitude below the data-side terms, indicating robustness against both discretionary cuts.

Behaviour of the inter-site contrast under independent Q_b revision. The inter-site contrast is far less sensitive to Q_b than the absolute K_d^* values, because a uniform rescaling $(K_d, Q_b) \to (\alpha K_d, \alpha Q_b)$ cancels exactly in ΔK_d^*; the contrast is exposed to Q_b uncertainty only through the differential between the two sites' rescaling factors. We test this explicitly by rescaling Q_b independently at the two sites across the full published envelope ($Q_{b,\mathrm{A15}} \in [14, 25]$, $Q_{b,\mathrm{A17}} \in [10, 18]$ mW m^{-2}; Fig. 9a). Along the global-rescaling diagonal the contrast scales proportionally and its significance in σ units is preserved. Off the diagonal the picture is sharper on one side and weaker on the other: the favourable extreme (A15 at the bottom of its Q_b envelope, A17 at the top) expands the contrast to ∼6.7 mW m^{-1} K^{-1}, while the most adverse corner (A15 at 25, A17 at 10 mW m^{-2}) collapses it to ≈ 0 — at that corner the two sites are consistent with a common conductivity. A −33% revision of $Q_{b,\mathrm{A17}}$ alone reduces the contrast to ∼0.8 mW m^{-1} K^{-1} ($\lesssim 1\sigma$). The conductivity contrast is therefore significant at the published basal fluxes ($p \approx 0.011$) and across moderate Q_b revisions, but it is *not* immune to the most adverse independent revision of both fluxes; what survives every admissible assignment is the difference in the constrained products Q_b/K_d themselves, which differ by a factor of ∼2.5 between the sites at the published fluxes.

A joint (K_d, H) fit over the swept $H = 3--10(A15) and 3--14(A17)$ cm range (Fig. 9b,c) confirms that H is effectively unconstrained: K_d and H trade off along a shallow valley whose interior minima, at $(4.3\,\mathrm{mW\,m^{-1}\,K^{-1}}, 8\,\mathrm{cm})$ for A15 and $(14.0, 9.3\,\mathrm{cm})$ for A17, improve on the 1-parameter retrieval at the published $H = 6$ cm by only 0.06 K of RMSE at both sites; the reported K_d^* is therefore conditional on the published H, and the K_d–H trade-off is propagated into the error budget as σ_H (Table 4). K_d, not H, is the identified parameterat fixed H; the pair is jointly identified only up to the valley direction.

4.3 Comparison with Martínez & Siegler (2021) and prior estimates

The second globally calibrated form fares notably better. The Martínez and Siegler (2021) $K(T, \rho)$ model, with its published coefficients and no per-site freedom, yields a meter-scale-sensor RMSE of 0.99 K at A15 and 0.64 K at A17 (Table 2, Fig. 8) — comparable to the Hayne site-fit at A15 and within a factor of ∼1.9 of it at A17. Its explicitly density- and temperature-dependent decomposition evidently transfers from the diurnal-skin calibration to the meter-scale regolith more faithfully than the Hayne smooth-exponential at its global K_d. The per-site Hayne retrieval nonetheless remains the sharper instrument at A17, where it halves the residual RMSE.

Diagnostic: Martinez per-site density retrieval. The fairness of the forward Martinez comparison can be tested by granting the published $K(T, \rho)$ one per-site lever and asking whether that lever can absorb the A17 residual within physical bounds. We note at the outset that Martínez and Siegler (2021) calibrated their four coefficients against surface brightness-temperature observations and did not validate the extrapolation to depths of order one meter; the test we run here is therefore a check of whether the published surface-calibrated form, when extended to the meter-scale regolith, can fit the in-situ HFE data with one per-site lever (the asymptotic bulk density entering the model). We do so under the construction of §2.5 (eq. 9): rescale the asymptote of the Hayne-form $\rho(z)$ entering equation (8) by a single per-site scalar α, leaving the four Martinez coefficients fixed, and sweep α against the same meter-scale-sensor RMSE. The retrieved values are $\alpha^*_{\mathrm{A15}} = 1.13$ (asymptotic bulk density $\rho_d^* \approx 2026$ kg m^{-3}, RMSE 0.90 K) and $\alpha^*_{\mathrm{A17}} = 0.98$ ($\rho_d^* \approx 1767$ kg m^{-3}, RMSE 0.63 K); at A15 the density lever matches the Hayne site-fit RMSE, while at A17 the bowl bottoms essentially at the published density and the lever buys almost nothing (0.64 → 0.63 K). The physical interpretation of α is the asymptotic bulk density that the published Martinez $K(T, \rho)$ would require to reproduce each site's in-situ profile.

The A15 retrieval sits marginally above the Apollo-core compaction ceiling of ∼2000 kg m^{-3} (Mitchell et al., 1973; Carrier et al., 1991) but remains physically plausible. The A17 retrieval bottoms at $\rho_d^* \approx 1767$ kg m^{-3}, comfortably inside the Apollo-core envelope (1700–2000 kg m^{-3}): the published $K(T, \rho)$ is physically admissible at both boreholes. Figure 10 visualises the result — both RMSE(α) bowls bottom inside or at the edge of the admissible band. The flat A17 bowl also exposes the limitation: because the bowl bottoms at the published density, no admissible density adjustment closes the remaining factor-of-∼1.9 RMSE gap to the per-site Hayne fit; sharpening the A17 gradient

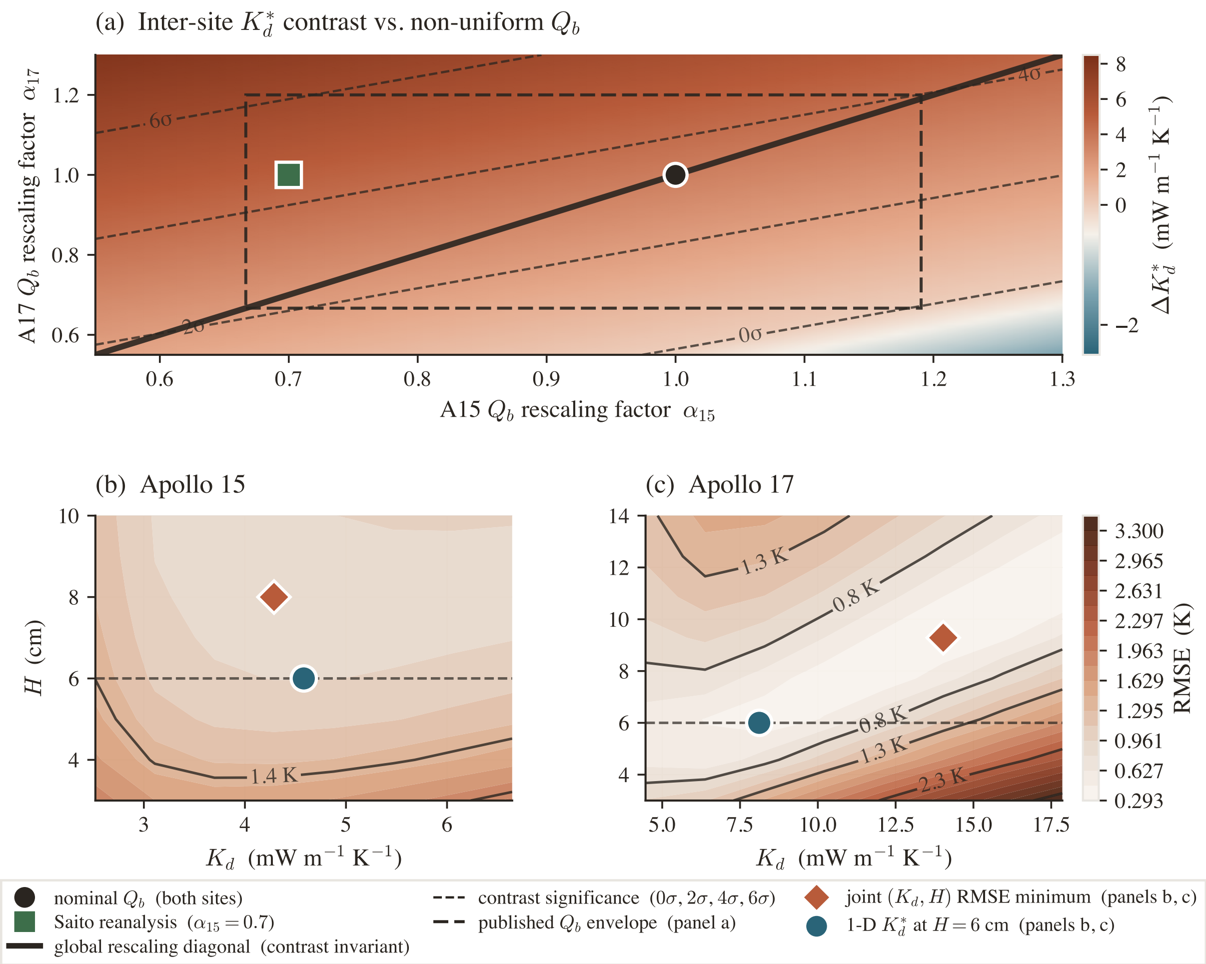

Figure 9. Robustness of the per-site K_d^* contrast. **(a)** Contrast ΔK_d^* as a function of per-site Q_b rescaling factors α_{15}, α_{17}. Solid diagonal: global rescaling; dashed contours: bootstrap-σ significance; black circle: nominal Q_b; green square: Saito et al. (2007) A15 reanalysis. **(b),(c)** Joint (K_d, H) RMSE surface at each site. Coral diamond: joint RMSE minimum; teal circle: 1-parameter retrieval at $H = 6$ cm.

further requires freedom in the conductivity mechanism itself (the contact/radiative split), not in the density input. A full per-site recalibration of the Martínez and Siegler (2021) coefficients against the HFE record is the natural follow-on study.

The inferred ratio $K_d^{\rm A17}/K_d^{\rm A15} \approx 1.8$ is physically plausible if interpreted as a conductivity effect. Two mechanisms act in the appropriate directionand jointly cover the required factor. First, Apollo cores span bulk porosities $\phi \sim 0.30$–0.45 (Mitchell et al., 1973; Carrier et al., 1991); the effective-medium scaling $K_{\rm eff} \propto (1-\phi)^2$ of Wood (2020) accounts for a factor of $\sim$1.5 change at the upper end of that range. Second, solid lunar basalt is $\sim$30% more conductive than solid highland anorthosite (Horai and Susaki, 1972), and A17 lies in a basaltic embayment while A15 occupies a mixed mare/highland margin (Korotev, 2005; Wieczorek et al., 2006; Crawford, 2015); the product of the two mechanisms ($\sim$1.9) matches the retrieved ratio without requiring either at its extreme.

Prior in-situ retrievals on the same HFE record provide a useful reference. The original Langseth et al. (1972, 1976) reductions yielded meter-scale conductivities of order 1–3 mW m^{-1} K^{-1} at both sites, derived from steady-state gradient/flux ratios under a layered-medium assumption that did not include the radiative-conductivity term embedded in modern $K(T, z)$ forms. The Keihm (1984) reanalysis and the Grott et al. (2010) critical reassessment both raise these values once the radiative term and the borestem heat-short are accounted for; the present $K_d^* = 4.58$ (A15) and 8.12 (A17) mW m^{-1} K^{-1} extend that upward trend, primarily because the entire upper 80 cm is excluded rather than corrected. Direct laboratory measurements on Apollo soils (Cremers and Birkebak, 1971; Horai, 1981; Hemingway et al., 1973) yield $\sim$0.7–2 mW m^{-1} K^{-1}, a factor of $\sim$3 below the A15 retrieval

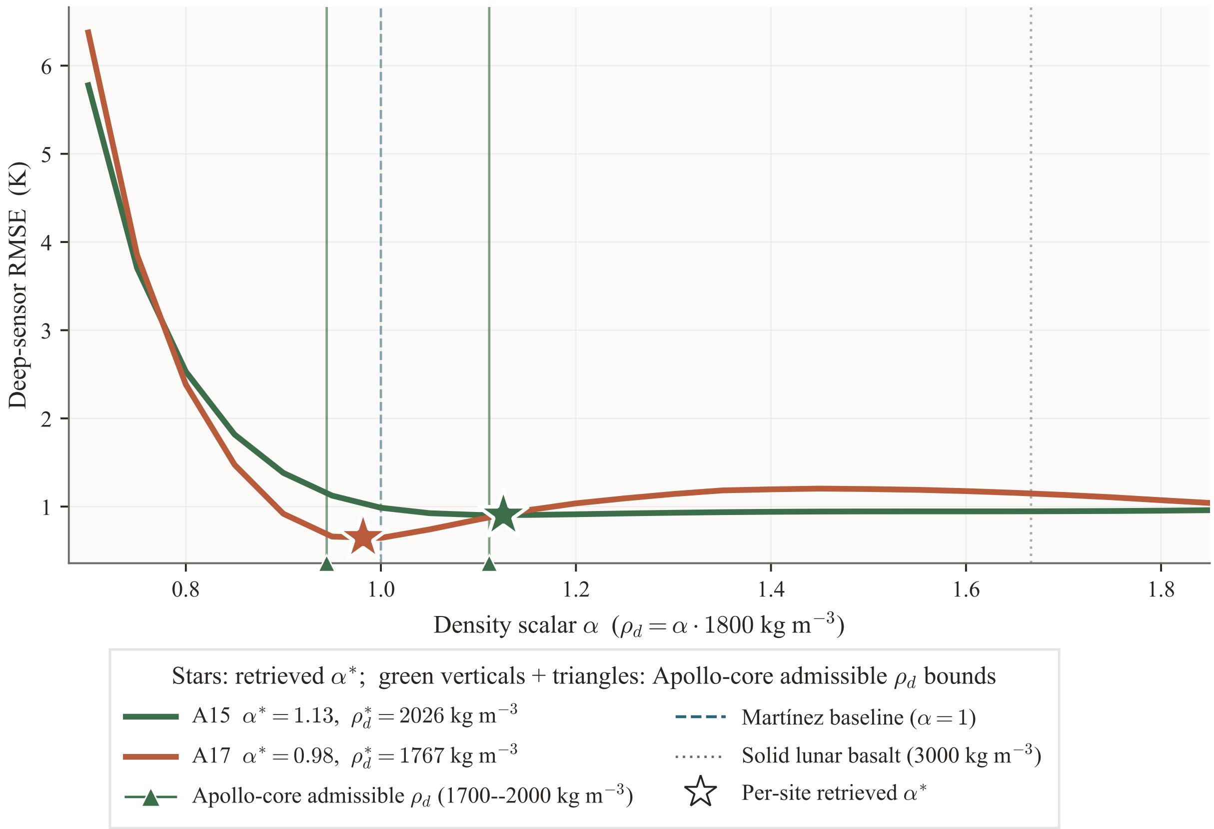

Figure 10. Diagnostic per-site retrieval under the Martínez and Siegler (2021) $K(T, \rho)$ form. Meter-scale-sensor RMSE as a function of the per-site density scalar α (eq. 9) for both sites, with the retrieved α^* starred. Dashed teal: published Martinez baseline ($\alpha = 1$, $\rho_d = 1800$ kg m^{-3}). Green verticals: Apollo-core admissible density range ($\rho_d = 1700$–2000 kg m^{-3}; Mitchell et al., 1973; Carrier et al., 1991). Dotted grey: solid-basalt upper bound (3000 kg m^{-3}; Wieczorek et al., 2006).

and $\sim$5 below A17, the expected lab-vs-in-situ systematic, in which laboratory samples lose the metre-scale grain-contact network and the radiative pathways that dominate the in-situ response (Wood, 2020; Feng et al., 2020). The present retrievals bracket the orbital Vasavada et al. (2012) estimate of $\sim$7 mW m^{-1} K^{-1}, with A15 below and A17 above.

Surface-temperature consistency check. The Diviner GCP is external to the retrieval (it enters neither the bootstrap nor the meter-scale-sensor RMSE), so it provides an independent check on the diurnal forcing chain (Fig. 11). Both parameterizations reproduce the canonical lunar diurnal shape, with full-cycle RMSE of 4.1 K at A15 and 7.3 K at A17 for the Hayne site fit, and 3.6 K at both sites for the Martinez forward. Both biases are dominated by the anisothermal-footprint and sub-pixel rock-population effects inherent to Diviner brightness temperatures (Bandfield et al., 2015; Hayne et al., 2017). Insolation, albedo, emissivity, and the upper-80 cm $K(T, z)$ are therefore correctly captured by both models, and the meter-scale-sensor discrepancy in Table 2 is not a surface-driven artefact.

At A17 the Hayne site fit runs surface-warmer than Diviner by +4.9 K while the Martinez forward sits at -0.8 K. The mechanism is direct: the high meter-scale $K_d^* = 8.12$ couples more heat upward through the diurnal skin layer than the globally calibrated $K(T, z)$ assumes, raising the dayside response. The per-site Hayne fit absorbs this penalty at the surface to fit the meter-scale temperature gradient; the surface-calibrated Martinez form remains essentially unbiased at the surface and pays a factor-of-$\sim$1.9 penalty in the meter-scale-sensor RMSE relative to the Hayne site fit (Table 2). Within the Hayne $K(T, z)$ family, no single parameter choice simultaneously optimises the surface-brightness constraint of Williams et al. (2017) and the meter-scale-gradient constraint of Nagihara et al. (2018) at A17; the $K(T, \rho)$ family narrows but does not close that gap. The inter-site difference flagged by the meter-scale retrieval reappears in the surface-temperature data, in the form of which model fits which site at the surface and by how much.

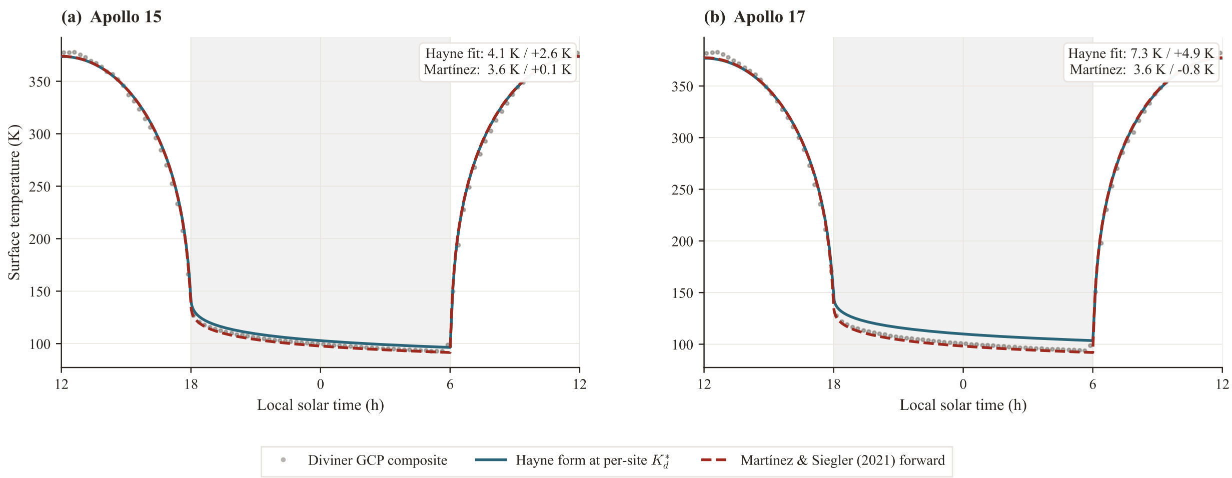

Figure 11. Surface-T closure against the Williams et al. (2017) Diviner GCP diurnal composite (grey markers). Teal solid: Hayne form at the per-site retrieved K_d^*; brick-red dashed: Martínez and Siegler (2021) form at published coefficients (no fitted knob). The local-solar-time axis is shifted so midnight sits at panel centre and noon at the edges; the shaded band (LST 18–06) marks the night side. Full-cycle RMSE / mean bias per panel are listed in the inset.

4.4 Physical plausibility of the retrieved K_d^*

Apollo 15. The retrieval gives $K_d^{\mathrm{A15}} = 4.58\,\mathrm{mW/(m\,K)}$. This value sits between the global Hayne et al. (2017) value (3.4) and the regional Vasavada et al. (2012) orbital estimate ($\sim$7), and extends the upward trend seen in the Keihm (1984) and Grott et al. (2010) reanalyses of the same HFE record once the radiative term and the borestem heat-short are corrected. Porosity provides the physical mechanism: the effective-medium scaling $K_{\mathrm{eff}} \propto (1-\phi)^2$ of Wood (2020), applied across the Apollo-core porosity envelope $\phi \sim 0.30$–0.45 (Mitchell et al., 1973; Carrier et al., 1991) and combined with the radiative term of the Hayne form, brackets $\sim$3–6 mW m^{-1} K^{-1}. The retrieved 4.58 falls inside that range. The factor-of-$\sim$3 gap below direct laboratory measurements (Cremers and Birkebak, 1971; Horai, 1981) reflects the standard lab-vs-in-situ systematic: laboratory samples lose the meter-scale grain-contact network and the radiative pathways that dominate the in-situ response. A15 is physically admissible.

Apollo 17. The retrieval gives $K_d^{\mathrm{A17}} = 8.12\,\mathrm{mW/(m\,K)}$, roughly 2.4 times the global Hayne value and 1.8 times the A15 retrieval. Two mechanisms acting together cover this range comfortably. First, solid lunar basalt is $\sim$30% more conductive than highland anorthosite (Horai and Susaki, 1972); A17 lies in a basaltic embayment of Mare Serenitatis while A15 occupies a mare/highland margin (Korotev, 2005; Wieczorek et al., 2006; Crawford, 2015). Second, the porosity-driven K scaling at the upper end of the Apollo-core range contributes a further $\sim 1.5\times$ enhancement (Wood, 2020). The product is $\sim 1.9\times$ the A15 value, matching the retrieval to within the bootstrap uncertainty. Neither contributor is required at its extreme, and the value sits just above the Vasavada et al. (2012) orbital estimate of $\sim$7.

A complementary test against the Martínez and Siegler (2021) $K(T,\rho)$ form (§4.3, Fig. 10) is consistent: the density that form requires at A17, $\rho_d^* \approx 1767$ kg m^{-3}, lies inside the Apollo-core envelope, independently corroborating that the A17 meter-scale column is unremarkable in density and that the enhanced conductivity is carried by composition and contact structure rather than by anomalous compaction.

The A17 absolute value is conditional on the adopted basal heat flux. The Saito et al. (2007); Nagihara et al. (2018) reanalyses revise Q_b^{A17} downward by up to $\sim$30%, which rescales the retrieved K_d^* to $\sim$5.4 mW m^{-1} K^{-1} through the K_d/Q_b degeneracy (§4.2). That rescaled value still exceeds the global value and falls within composition+porosity bounds. The factor-of-2.4 excess over the global Hayne value therefore cannot be read as a measured absolute conductivity independent of the

Q_b assumption.

A site–depth ambiguity also deserves explicit mention: the two sensor sets barely overlap in depth (80–139 cm at A15, 130–243 cm at A17), so a conductivity that continues to increase below ~1.5 m — continued compaction rather than a lateral site difference — could in principle mimic the contrast. Two internal diagnostics limit this alternative. Within the A17 set itself, which spans 1.1 m, a depth-constant K_d leaves only the 0.34 K residual RMSE of Table 2 and predicts the withheld deepest sensor (234 cm) to 0.09 K (§3.3); a $K(z)$ still rising across that span would curve the profile detectably. The borestem-cut sweep similarly leaves both retrievals unchanged as the A15 set is restricted to its deepest five sensors ($z_b = 90$ cm, §3.2). Depth-dependent compaction below the probed range cannot be excluded with two boreholes, but it is not required by the data, whereas the compositional difference between the sites is independently established.

A17 remains the more conductive site at the published basal heat fluxes and across moderate revisions; only the most adverse corner of the independent-Q_b envelope (§4.2) erases the contrast. Within every admissible combination of composition and porosity the sign is unchanged, even where the absolute A17 value is not.

4.5 Limitations and caveats

Five caveats accompany any use of the result. *(i)* With only two in-situ boreholes, the inter-site difference is insufficient to establish regional heterogeneity across the lunar surface; distinguishing genuine regional variability from local heterogeneity at two specific borehole locations requires additional landed thermometry. *(ii)* The meter-scale temperature gradient constrains the product Q_b/K_d, not K_d alone (§4.2); the data favor but do not establish a conductivity contrast over a uniform conductivity with differing basal flux. *(iii)* The contrast magnitude is significant at the 95% level only conditional on the published basal heat fluxes ($\Delta K_d^* = 3.3$, bootstrap CI $[0.4, 4.6]$, $p \approx 0.011$); propagating the Saito et al. (2007); Nagihara et al. (2018) Q_b reanalysis envelope dilutes the ordering probability to ~83%, and the most adverse independent Q_b assignment erases the contrast entirely. *(iv)* The retrieval adopts the Hayne et al. (2017) functional form for $K(T, z)$; alternative piecewise parameterizations are not exhausted, and a full per-site recalibration of the Martínez and Siegler (2021) coefficients (whose density lever is already admissible at both sites but saturates at A17, §4.3) is the immediate next step. *(v)* The diurnal-amplitude depth profile (Fig. 4) serves only as a borestem diagnostic and not as a frequency-domain validation, because below 80 cm both models predict amplitudes well below the 1971–1977 digitisation noise floor.

5 Conclusions

The two Apollo boreholes require distinctly different meter-scale conductivities. Holding the Hayne et al. (2017) functional form fixed and sweeping K_d alone against the meter-scale-sensor RMSE, the per-site retrieval yields $K^*_{d,\mathrm{A15}} = 4.58^{+1.33}_{-0.28}$ and $K^*_{d,\mathrm{A17}} = 8.12^{+0.49}_{-0.61}\,\mathrm{mW/(m\,K)}$ (1σ non-parametric bootstrap, $N_{\mathrm{boot}} = 1500$) at the published basal heat fluxes, with the inter-site contrast significant at the 95% level ($\Delta K_d^* = 3.3$, 95% CI $[0.4, 4.6]$, $p \approx 0.011$). The meter-scale-sensor RMSE decreases from 1.03 K and 0.69 K under the published global $K_d = 3.4\,\mathrm{mW/(m\,K)}$ to 0.95 K and 0.34 K under the per-site retrieval; the Martínez and Siegler (2021) $K(T, \rho)$ forward evaluation, calibrated against the same family of orbital products, reproduces both sites to within ~1 K without per-site freedom, though it cannot match the sharpened per-site A17 fit (Table 2). The inter-site requirement is robust to the borestem exclusion depth, the stability-window threshold, a joint K_d–H fit, and the independent Diviner surface-temperature closure check.

The absolute K_d^* values are conditional on the adopted Q_b. Because the meter-scale temperature gradient constrains the product Q_b/K_d rather than K_d alone, the inter-site difference admits two interpretations: a conductivity contrast or a basal-flux contrast. A pooled three-model comparison favors the conductivity reading (weakly, $\Delta\mathrm{AICc} = 2.7$–$4.0$; the per-site evidence at A17 is decisive, $\Delta\mathrm{AICc} \approx 19$) (Table 3) without excluding the alternative. The quantity constrained tightly at each borehole is the meter-scale temperature gradient itself.

The immediate application is to forward thermal models used by upcoming lunar sub-surface RTM retrievals. These retrievals require a meter-scale $T(z)$ boundary condition that orbital-only conductivity models cannot supply on a per-site basis. The per-site K_d^* values reported here, evaluated in the Hayne thermal column, provide that boundary condition (§1); the upcoming TSUKIMI terahertz mission (Kasai et al., 2022) is one immediate beneficiary. A full per-site recalibration of the Martínez and Siegler (2021) form, and the extension of the meter-scale $T(z)$ column to include surface-shadowing effects on the RTM side, are both deferred to follow-on work. Extending the analysis beyond two boreholes, to a regionally resolved treatment of the meter-scale lunar thermal state, is fundamentally a measurement problem that requires meter-scale temperature data from additional landed instruments (e.g. Chang'E and ChaSTE-class experiments).

Acknowledgments

The authors thank the original Apollo HFE Principal Investigators and the Nagihara et al. (2018) team for restoring the 1975–1977 portion of the 1971–1977 record from the original tapes (the 1971–1975 portion was retained from the original PI processing), without which this work would not have been possible.

Open Research

Availability Statement. The restored Apollo 15 and 17 Heat-Flow Experiment temperature record (1971–1977) analyzed here is openly available as the supporting information of Nagihara et al. (2018) (`https://doi.org/10.1029/2018JE005579`). The Diviner Lunar Radiometer Global Cumulative Product used for the out-of-sample surface-temperature closure (Williams et al., 2017) is available from the NASA Planetary Data System Geosciences Node (`https://pds-geosciences.wustl.edu`; LRO-L-DLRE-5-GCP-V1.0, `https://doi.org/10.17189/1520650`). The repository bundles the NASA SPICE Toolkit kernels (JPL DE440; Acton, 1996; Acton et al., 2018) used by its ephemeris module. The 1-D Crank–Nicolson heat solver, the ephemeris-driven solar-forcing routines, the K_d retrieval and bootstrap code, the validation diagnostics, and the data and scripts needed to reproduce every figure and table in this manuscript are archived at `https://github.com/rp3gregorio/Lunar-HFE`; a Zenodo-archived release with a citable DOI will be issued at the time of acceptance and added to the reference list.

Conflict of Interest Disclosure

The authors declare there are no conflicts of interest for this manuscript.

Nomenclature

Symbols.

Symbol	Definition	Value / units	Source
K_d	meter-scale regolith thermal conductivity (retrieved)	4.58, 8.12 $\mathrm{mW\,m^{-1}\,K^{-1}}$	this work
K_s	surface regolith thermal conductivity	7.4×10^{-4} $\mathrm{W\,m^{-1}\,K^{-1}}$	Hayne et al. (2017)
H	e-folding depth of $K(z)$ transition	0.06 m	Hayne et al. (2017)
χ	radiative conductivity multiplier coefficient	2.7	Hayne et al. (2017)
T_{ref}	normalisation temperature for radiative term	350 K	Hayne et al. (2017)
ρ_s, ρ_d	surface / asymptotic regolith bulk density	1100, 1800 $\mathrm{kg\,m^{-3}}$	Hayne et al. (2017)
$c_p(T)$	specific heat (4th-order polynomial in T)	$\mathrm{J\,kg^{-1}\,K^{-1}}$	Hemingway et al. (1973); Hayne et al. (2017)
Q_b	basal heat flux	21 (A15), 15 (A17) $\mathrm{mW\,m^{-2}}$	Langseth et al. (1976); Nagihara et al. (2018)
S_0	solar constant (irradiance at 1 AU)	1361 $\mathrm{W\,m^{-2}}$	Kopp and Lean (2011)
A, ε	broadband Bond albedo, emissivity	site-specific; 0.95	Vasavada et al. (2012); Bandfield et al. (2015)
σ	Stefan–Boltzmann constant	5.6704×10^{-8} $\mathrm{W\,m^{-2}\,K^{-4}}$	CODATA 2018
$\theta_\odot(t)$	solar zenith angle at site	rad	synodic approximation, §2
$F_\odot(t)$	incident solar irradiance at surface	$\mathrm{W\,m^{-2}}$	this work, eq. 1
$T(z,t)$, T_s	subsurface and skin ($z=0$) temperature	K	this work
z	depth coordinate (positive downward)	m	—
80 cm	borestem-exclusion depth (*a priori* cut)	80 cm	this work, §2.4
δ	diurnal thermal skin depth	∼3–5 cm	Hayne et al. (2017)
ΔK_d	inter-site contrast $K_d^{\mathrm{A17}} - K_d^{\mathrm{A15}}$	3.31 $\mathrm{mW\,m^{-1}\,K^{-1}}$	this work
RMSE	meter-scale-sensor root-mean-square residual	K	this work
N	number of meter-scale sensors	7 (A15), 16 (A17)	Nagihara et al. (2018)
N_{boot}	non-parametric bootstrap resamples	1500	Efron and Tibshirani (1993)
p	bootstrap proportion below null contrast	0.011	this work

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621 *Subscripts.*

s	surface (regolith, skin)
d	asymptotic (deep / meter-scale regolith)
b	basal (lower BC)
ref	reference (normalisation)
$\odot$	solar
eq	stability-window equilibrium
boot	bootstrap quantity

Superscripts.

*	best-fit / retrieved (e.g. K_d^*)
$^{\mathrm{A15,A17}}$	per-site qualifier
$^{\mathrm{ref}}$	published reference value
$^{\mathrm{HFE}}$	Apollo HFE-measured
$\langle\cdot\rangle$	time-average over stability window

622 *Acronyms.*

HFE	Heat-Flow Experiment (Apollo 15 and 17 buried thermometers)
TG / TR	gradient-bridge / ring-bridge sensor (HFE sensor types)
LST	local solar time
Diviner GCP	Diviner Lunar Radiometer Global Cumulative Product
MCMC	Markov-chain Monte Carlo
BCa	bias-corrected accelerated bootstrap
SPICE	NASA ancillary information system for planetary geometry
DE440	JPL planetary and lunar ephemeris release 440
RTM	radiative-transfer model
TSUKIMI	lunar mission consuming the meter-scale $T(z)$ retrieved here

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